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VALIDATION REPORT

ATMOSPHERIC CORRECTION FOR SENTINEL-3 OLCI AND SLSTR PRODUCTS

VERSION 1.1

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List of Acronyms

6S	Second Simulation of the Satellite Signal in the Solar Spectrum
AC	Atmospheric Corrections
ACIX	Atmospheric Corrections Intercomparison eXercice
AD	Applicable Document
ADAS	Atmospheric Data Assimilation System
AERONET	Aerosol Robotic Network
AOT	Aerosol Optical Thickness
ATBD	Algorithm Theoretical Basis Document
CAMS	Copernicus Atmosphere Monitoring Service
CGLOPS	Copernicus Global Land Operations
FWHM	Full Width at Half Maximum
GEOS	Goddard Earth Observing System
GTOPO30	Global 30 arc s elevation dataset
HYGEOS	HYdro-GEologie et Observation Spatiale
MERRA-2	Modern-Era Retrospective analysis for Research and Applications, Version 2
NASA	National Aeronautics and Space Agency
NetCDF	Network Common Data Form
NRT	Near Real Time
OLCI	Ocean and Land Colour Instrument
OPAC	Optical Properties of Aerosols and Clouds
PROBA-V	Project for On Board Autonomy - Vegetation
QAR	Quality Assessment Report
RT	Radiative Transfer
S3	Sentinel-3
SLSTR	Sea and Land Surface Temperature Radiometer
SMAC	Simplified Model for Atmospheric Corrections
SSA	Single Scattering Albedo
SSD	Service Specifications Document
SVP	Service Validation Plan
SZA	Solar Zenith Angle
TOA	Top Of the Atmosphere
TOA-r	Top Of Atmosphere Reflectance
TOC	Top Of Canopy
TOC-r	Top Of Canopy Reflectance
UTC	Universal Time Coordinates

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series.

CLMS portfolio contains a number of global "Vegetation" biophysical products based upon the data acquired by the Ocean and Land Colour Instrument (OLCI) and the Sea and Land Surface Temperature Radiometer (SLSTR) onboard the Sentinel-3 (S3) platform. All various biophysical algorithms ingest the same surface reflectances derived from the OLCI and SLSTR Level 1B data by a harmonized pre-processing chain.

This report presents the methodology and the results of the analysis carried out to assess the performance of the atmospheric correction scheme applied on Sentinel-3 OLCI and SLSTR Top-Of-Atmosphere (TOA) reflectances. The methodology follows the approach defined by Doxani et al, 2018 and applied in the Atmospheric Correction Inter-Comparison Exercise "ACIX". The satellite Top-Of- Canopy (TOC) reflectances are compared to a reference product retrieved using an accurate radiative transfer code and inversion products of the in-situ Aerosol Robotic Network (AERONET) for aerosols optical thickness and model. Because the quality of the TOC reflectance depends on the quality of the ancillary data, in particular on aerosols load, three sources of ancillary data are investigated: MERRA-2 daily reanalysis, NRT products from the Copernicus Atmosphere Monitoring Service (CAMS) and a monthly climatology. Performances of atmospheric corrections using CAMS NRT and MERRA-2 reanalysis are very similar. The use of a climatology gives degraded results. For the domain of application of the AC scheme, with a filter on high values of aerosol optical thickness, CAMS NRT dataset is suitable for the operational production of the Sentinel-3 TOC reflectances, further used as inputs for the production of biophysical products. Furthermore, comparison of TOC-reflectance products, generated by the CLMS Global Land Operations (CGLOPS-1) production lines in version 1 and version 2, show good agreements despite of different period of observations. The analysis of the error bars on the TOC reflectances shows the advantage of the version 2 because the uncertainty due to the aerosol model is included in the error bar and is globally of the same order of magnitude than the ACIX uncertainty.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document presents the evaluation of the performance of the atmospheric correction scheme applied in the TOA-to-TOC pre-processing chain of the Sentinel-3 OLCI and SLSTR sensors data.

The analysis is performed on one year of (i) TOC reflectances version 1 (July 2018-June 2019 for Sentinel-3A and January to June 2019 for Sentinel-3B) and (ii) TOC reflectances version 2.0 (year 2020 of Sentinel-3A and 3B), both retrieved over AERONET sites.

Note: the results obtained for the TOC reflectances version 2.0 are also valid for the TOC reflectances version 2.3 (see section 4.1 for details).

The objective is to check that the atmospheric correction performs well and, thus, that the retrieved TOC reflectances are suitable for the production of the biophysical products.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 describes the methodology, the metrics and the criteria of evaluation
- Chapter 3 presents the results of the performance analysis of the atmospheric correction
- Chapter 4 addresses the evaluation of the TOC reflectances, the evaluation of their error bars and summarizes the main conclusions of the study
- Chapter 5 makes recommendations based upon the results

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document for Vegetation & energy products of the Global Land Service
CGLOPS1_ATBD_S3-AC-V1.1	Algorithm Theoretical Basis Document of the atmospheric correction Version 1.1 applied on Sentinel-3 OLCI and SLSTR products

Available at https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Validation report of the CAMS near-real-time global atmospheric composition service: Period September-November 2018, Copernicus Atmosphere Monitoring Service (CAMS) report, CAMS84_2018SC1_D1.1.1_SON2018_v1.pdf, March 2019, doi:10.24380/dg9cpm41.

http://atmosphere.copernicus.eu/quarterly_validation_reports

S3MPC.RAL.TN.020	Sentinel-3 SLSTR VIS and SWIR Channel Vicarious Calibration adjustments, Issue 3.0, 22.05.2021.

https://www-cdn.eumetsat.int/files/2022-10/S3MPC_VIS_SWIR_Vicarious_Calibration_Adjustments%203.0.pdf

2 QUALITY ASSESSMENT METHOD

2.1 OVERALL PROCEDURE

The Simplified Model for Atmospheric Corrections (SMAC) algorithm (Rahman and Dedieu, 1994) is used to perform atmospheric correction in the TOA-to-TOC module of the CGLS Sentinel-3 OLCI and SLSTR biophysical processing chains. It is chosen for its speed, robustness and simplicity. Inputs like the Aerosol Optical Thickness (AOT), the total column of ozone, the amount of water vapour and the surface pressure are from ancillary data. The aerosol model is assumed to be the "continental" one. Details are presented in the ATBD [CGLOPS1_ATBD_S3-AC-V1.1].

Part of the uncertainty and errors on the TOC reflectance (TOC-r) is due to the quality of the ancillary data used in the atmospheric correction scheme. Thus, the sensitivity of the TOC-r on source of ancillary data, especially aerosols, is investigated using different sources to perform atmospheric correction (see Section 2.2.1). Moreover, the different AOT datasets are inter-compared for each match-ups with the in-situ AERONET reference product over more than 40 stations during one year.

The performance of the atmospheric corrections scheme is evaluated through the assessment of the quality of the TOC-r. The analysis is carried out by comparing TOC-r with in-situ reference product obtained from AERONET data and the Second Simulation of the Satellite Signal in the Solar Spectrum (6S) accurate atmospheric correction method. Thus, it follows clearly the recommendation of the Atmospheric Correction Inter-Comparison Exercise "ACIX" (Doxani et al, 2018). The comparison is performed for more than 40 AERONET sites that cover different biomes, localisations, meteorological and states of the atmosphere.

The overall methodology, including atmospheric corrections methods and used ancillary data, is summarized in Figure 1.

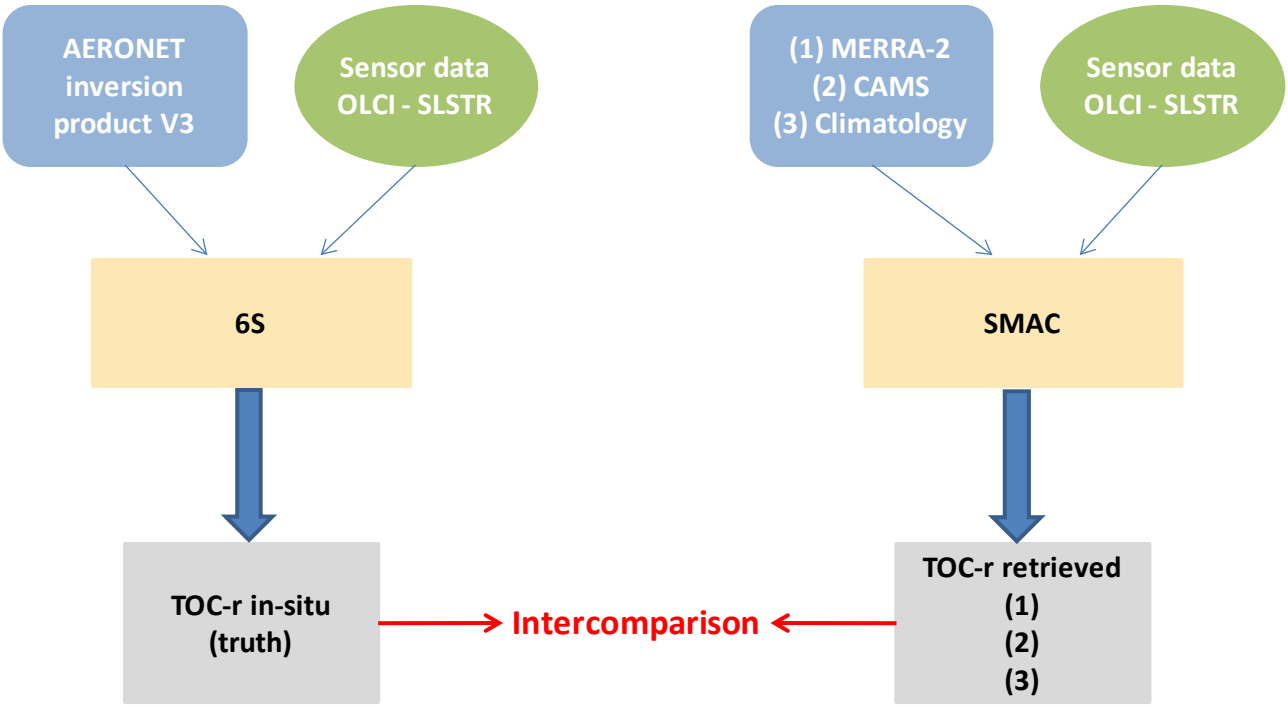


Figure 1: Flowchart of the methodology to evaluate the performance of atmospheric correction scheme and to assess the impact of ancillary data.

The criteria of performance and quality are based on metrics to quantify agreement between the satellite TOC-r product and the in-situ reference products. The mean bias, the standard deviation of bias and the root mean square deviation (RMSD) are the mathematical expressions of ACIX metrics accuracy (A), precision (P) and uncertainty (U), respectively. These quantities are given in formula [1], [2] and [3].

$$A = \frac{1}{N}\left(\sum_{i=1}^{N} \Delta toc_r_i\right) \qquad \text{Eq.1}$$

$$P = \sqrt{\frac{1}{N-1}\left(\sum_{i=1}^{N} (\Delta toc_r_i - A)^2\right)} \qquad \text{Eq.2}$$

$$U = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\Delta toc_r_i)^2} \qquad \text{Eq.3}$$

where

$$\Delta toc_r_i = \left(toc_r_i^{processor} - toc_r_i^{Aeronet}\right) \qquad \text{Eq.4}$$

are the residuals between the satellite product $toc_r_i^{processor}$, i.e. the TOC reflectance calculated by the SMAC processor using ancillary data (left branch of Figure 1), and the "truth", $toc_r_i^{Aeronet}$, i.e. the TOC reflectance calculated using 6S and AERONET inversion product for the AOT and the aerosol model (right branch of Figure 1). N is the number of match-ups.

Metrics are summarized in Table 1.

Table 1: Metrics used for the validation to assess the quality of the satellite product.

Metrics	Comment
N: Number of samples	Indicative of the power of the validation
Uncertainty U **Root Mean Square Deviation**	RMSD is the square root of the average of squared errors between x and y. Indicates the total error or uncertainty.
Accuracy A **Mean Bias**	Difference between average values of x and y. Indicative of accuracy and possible offset.
Precision P **Standard deviation of bias**	Standard deviation of the pair differences. Indicates precision.
R: Correlation coefficient	Indicates descriptive power of the linear accuracy test.
Parameter of the linear regression: slope and offset	Indicates some possible and systematic bias.

Total measurement uncertainty (U, i.e., root mean square deviation) includes systematic measurement error (A, i.e., Bias) and random measurement error (P, i.e., Standard deviation of bias). RMSD is recommended as the overall performance statistic. Least square regression is also used to quantify the goodness of the fit: intercept and slope are other good indicators of possible bias. Finally, the

correlation coefficient of the least square regression is the last metrics to assess the agreement between the satellite product and the in-situ product.

2.2 SENTINEL-3 OLCI AND SLSTR TOC REFLECTANCES

The TOC-r is derived from Sentinel-3 OLCI and SLSTR sensors measurements through atmospheric correction using as inputs the TOA reflectances, the solar and view angles and ancillary data characterizing the surface and atmosphere (surface elevation, water vapor, pressure, aerosol, ozone). About the aerosol's information, only the AOT is given by external ancillary data; the aerosol model is assumed to be the "continental" one.

Table 4 and Table 5 in Annex 1: Spectral bands of Sentinel-3 sensors summarize the spectral bands of OLCI and SLSTR, respectively, for which TOC-r are calculated.

2.2.1 Ancillary data

To perform accurate atmospheric correction, the state of the atmosphere must be known. Relevant parameters that describe the atmosphere are coming from ancillary data. Because the quality of the TOC-r product depends on the quality of the ancillary data, in particular on aerosols data, three sources of ancillary data are investigated.

2.2.1.1 MERRA-2 dataset

The Modern-Era Retrospective analysis for Research and Applications version 2 (MERRA-2) is a NASA atmospheric reanalysis for the satellite era using the Goddard Earth Observing System Model, Version 5 (GEOS-5) with its Atmospheric Data Assimilation System (ADAS), version 5.12.4. It is the data source for all atmospheric parameters (https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/). The README file for the NetCDF MERRA-2 products is available here: (https://goldsmr4.gesdisc.eosdis.nasa.gov/data/MERRA2/M2I1NXINT.5.12.4/doc/MERRA2.README.pdf), while the full product specification is available here: (https://gmao.gsfc.nasa.gov/pubs/docs/Bosilovich785.pdf).

The datasets are available from 1st January 1980 until present. They can be downloaded from several sites, for example (https://disc.sci.gsfc.nasa.gov/daac-bin/FTPSubset2.pl).

Two products are currently used for all atmospheric parameters:

1. `tavg1_2d_slv_Nx`: 2-dimensional, 1-Hourly, Time-Averaged for physical parameters, ozone and water vapour.
2. `tavg1_2d_aer_Nx`: 2-dimensional, 1-Hourly, Time-Averaged for aerosol diagnostics.

Both of them have the same spatial and temporal grids:

- Frequency: 24 1-hourly from 00:30 UTC (time averaged)
- Spatial Grid: 2D, single-level, global Lat-Lon regular grid
- Lat: 361 values from -90 to 90
- Lon: 576 values from -180 to 179.375

The spatial resolution of MERRA-2 is about 50 km in the latitudinal directional. It is relatively coarse but MERRA-2 data are spatially interpolated to remove possible hard spatial transition due, for example, to two cells with extreme values.

MERRA-2 dataset is used in this analysis because it is a well-known source of ancillary data. Since it covers the long time period, it has been used for archive processing in the Copernicus Climate Change Service (C3S) for generating land products like the albedo, the LAI and the FAPAR.

2.2.1.2 CAMS NRT dataset

The Copernicus Atmosphere Monitoring Service (CAMS) Near Real Time (NRT) data are 3-hourly product (Inness et al., 2019). The CAMS NRT data are daily retrieved as grib files, through ftp download from:

dissemination.ecmwf.int /DATA/CAMS_GLOBAL/<YYYYMMDD>00, with <YYYYMMDD> being the date of interest.

Products of forecast time 0:00 and in steps of 3h are downloaded.

The CAMS NRT dataset is used for the CGLS operations on Sentinel-3 data. They are spatially and temporally interpolated to match with the Sentinel-3 overpass time.

2.2.1.3 A climatology

A climatology based upon a MERRA_2 monthly synthesis of 2018 is included into the analysis. It can also be downloaded at:

https://disc.gsfc.nasa.gov/datasets?keywords=%22MERRA-2%22&page=1&source=Models%2FAnalyses%20MERRA-2

- `tavgM_2d_slv_Nx`: 2-dimensional, Monthly Mean, Time-Averaged for physical parameters, ozone and water vapour.
- `tavgM_2d_aer_Nx`: 2-dimensional, Monthly Mean, Time-Averaged for aerosol diagnostics.

In case of non-availability of nominal ancillary dataset (CAMS NRT product), this climatology is used as back-up for the CGLS operations on Sentinel-3 data.

2.2.1.4 Accuracy of ancillary data

Table 2 summarizes the information on the accuracy of the MERRA-2 and CAMS ancillary datasets. Due to the lack of literature on water vapour uncertainty, it has been decided to assign an error of 20% on its parameter. Accuracy is not given for the monthly climatology because of the temporal variability: within a month, day to day variability is higher than the accuracy of the monthly mean.

Table 2: Accuracy, in terms of uncertainty, of ancillary data

Product	MERRA-2 (1-hourly resolution)	CAMS (3-hourly resolution)
O_3	Less than 6% *(Wargan et al., 2017, Davis et al., 2017)*	*5% for stratospheric ozone, 10% for tropospheric ozone [CAMS84_2018SC1_D1.1.1_SON2018_v1]*
AOT	RMS of 0.13 *(Randles et al., 2017)*	0.040 ±0.194 (Statistics over 48 AERONET stations).
Surface pressure	Less than 1 hPa *(van den Besselaar et al., 2011)*	Less than 1 hPa *(van den Besselaar et al., 2011)*

2.2.2 Summary

The satellite TOC-r products ($toc_r_i^{processor}$ in Eq.4) are retrieved as follow (see [CGLOPS1_ATBD_S3-AC-V1.1] for more details):

- Inputs:
  - Sun-view geometry, latitude, longitude, time and TOA-r from OLCI and SLSTR sensors
  - AOT, ozone total column, water vapour column and surface pressure from
    i. MERRA-2 1-hourly product or
    ii. CAMS 3-hourly product or
    iii. A climatology

    - the aerosol "continental" model is assumed
- AC scheme: SMAC

Two versions, V1.0 and V1.1, of the atmospheric corrections algorithm are considered. The core of the algorithm is the same except one change in V1.1: the uncertainty on TOC reflectances due to the aerosol model is taken into account in the calculation of the error on TOC reflectance.

The outputs of the atmospheric corrections algorithm V1.0 are the TOC reflectances version 1.

The outputs of the atmospheric corrections algorithm V1.1 are the TOC reflectances version 2.0 and greater.

2.3 IN-SITU REFERENCE PRODUCTS

AERONET inversion products Version 3 are used as in-situ aerosols measurements (Holben et al., 2006; Sinyuk et al., 2020, see also AERONET webpage https://aeronet.gsfc.nasa.gov/new_web/Documents/Inversion_products_for_V3). 47 AERONET sites around the world (Table 6 in Annex 2: AERONET sites) cover various types of surfaces, atmospheric states and continents. The relevant AERONET aerosols inversion products are:

- Aerosol Optical Thickness
- Index of refraction (real and imaginary parts) at different wavelengths
- The volume particle size distribution

These products are assumed to be in-situ truth information on aerosol load and granulometry and allow characterizing well the radiative and optical properties of aerosols in radiative transfer (RT) codes. Using AERONET inversion products allows us to have a large set of aerosols size distribution and index of refraction, and thus it allows us to have a very large set of aerosols models using a representative set of AERONET sites. Ozone content, water vapour content and surface pressure can also be present in AERONET data depending on the date and the site. If these three parameters are not present in AERONET files, daily MERRA-2 reanalysis will be used. Aerosols products, total column ozone and water vapour, surface pressure are used as inputs to the RT code 6S (Vermote et al., 1997).

The in-situ reference product, so called TOC-r-6S-AERONET ($toc_r_i^{Aeronet}$ in Eq.4), is the Top of Canopy reflectance simulated and retrieved by 6S using TOA-r from OLCI and SLSTR and AERONET inversion products.

Bias and uncertainty estimates for near real time AOT are computed by using the difference of the pre-field calibration AOT minus the interpolated calibration AOT (Giles et al, 2019). The near real time AERONET data have an estimated bias of +0.02 and 1σ uncertainty of up to 0.02: these values

have slightly higher uncertainty for shorter wavelengths and slightly lower uncertainty for longer wavelengths (Giles et al., 2019). Uncertainties can also vary depending on the model and the size of aerosols.

To summarize, the in-situ TOC-r-6S-AERONET is retrieved as follows:

- Inputs:
  - AOT, aerosols model, total column ozone, water vapour content and surface pressure from AERONET inversion product Version 3.
- AC scheme: 6S

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3 RESULTS

3.1 COMPARISON OF AEROSOL OPTICAL THICKNESS DATASETS

In a first step, the AOT coming from MERRA-2, CAMS, and the climatology, used as input in the SMAC processor, are compared to the AERONET level 1.5 AOT at 550 nm from the inversion product version V3. Only AERONET measurements within a ±1 hour time difference from the Sentinel-3 overpass are kept to avoid large temporal variation of the AOT. Results of the intercomparison are shown in Figure 2. 1845 match-ups are obtained. Best agreement with AERONET data is found for the MERRA-2 AOT product. For example, the smallest bias, on absolute value, is obtained with the MERRA-2 product (-0.0098). A bias of 0.0183 is found with CAMS NRT whereas a bias of 0.0359 is found for the climatology. No values of AOT higher than 1.0 is found for the climatology.

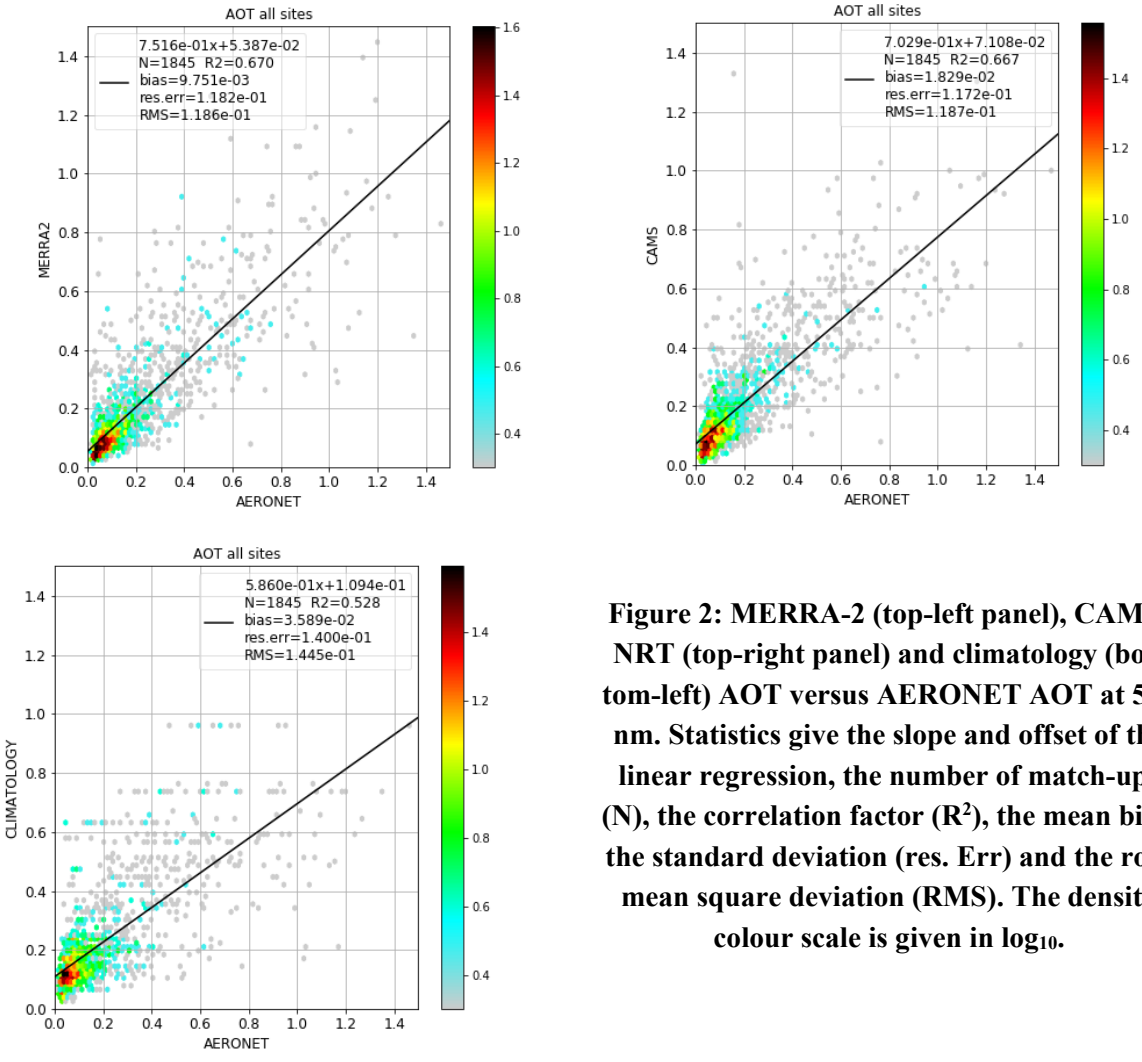


Figure 2: MERRA-2 (top-left panel), CAMS-NRT (top-right panel) and climatology (bottom-left) AOT versus AERONET AOT at 550 nm. Statistics give the slope and offset of the linear regression, the number of match-ups (N), the correlation factor (R^2), the mean bias, the standard deviation (res. Err) and the root mean square deviation (RMS). The density colour scale is given in $\log_{10}$.

3.2 EVALUATION OF AC SCHEME PERFORMANCE

3.2.1 Surface Reflectance from OLCI instrument

3.2.1.1 Statistics for best cases

For the validation of the AC scheme, we filter the data, keeping the pixels as following:

- Cloud free pixels not contaminated by cloud shadows
- Pixels corresponding to moderate solar or viewing zenith angles (SZA<65° and VZA<65°). For S3 SLSTR and OLCI, VZA never reaches this value. SZA can be greater than 65° depending on the latitude and the season. This value is higher than the value of 60° suggested by Rahman and Dedieu (1994) because radiative transfer calculations are now more accurate and better fits of radiative transfer calculations allow more accurate retrieval of SMAC coefficients.
- Pixels not contaminated by high aerosol load (AOT<1.)

The first criterion is obvious. To identify clear-sky pixels, we use the information of the SLSTR cloud mask (cloud_an) and the OLCI IdePix flag (pixel_classif_flags).

The second criterion is due to limitations in radiative transfer calculation. Indeed, the higher the zenith angles, the higher is the atmospheric airmass: the atmospheric contribution to the TOA reflectance increases significantly. For grazing zenith angles, the distance between the top of the atmosphere can be very large and the spatial variability of the atmospheric characteristics must be taken into account. Moreover, the plane parallel assumption used in radiative transfer calculations is not accurate anymore: for angles greater than 70 degrees, a spherical shell atmosphere must be considered. Finally, the SMAC method is based on fit on radiative transfer simulation to model the signal into its different components (gaseous absorption, aerosols and molecules scattering, surface, interaction between surface and atmosphere). These fits are optimized for angles less than 60 degrees. The accuracy of the SMAC method decreases then after this value.

The last criterion is based on the load of aerosols in the atmosphere. The higher is the AOT, the larger is the quantity to be corrected to obtain the surface reflectance from the TOA reflectance. As for the zenith angles, SMAC accuracy starts decreasing when AOT becomes greater than 0.8 (Rahman and Dedieu, 1994).

Over the whole period (July 2018-June 2019) and the 48 AERONET sites, the resulting number of match-up is about 1700. Only five negative TOC reflectances are found for the OLCI band Oa02 obtained with MERRA-2. No negative values are found for the other bands. Results of the comparison between the AC processor and 6S using AERONET are given in Figure 3 on accuracy, Figure 4 on precision and Figure 5 on uncertainty.

The accuracy is typically around 0.006 on average with variations depending on the spectral bands. Accuracy is better at shorter wavelength, in the blue for example, with value around 0.002 for band Oa03 (centred at 443 nm), and is worse for Oa11 (centred at 709 nm) with values around 0.010. Accuracy, or the bias, is systematically positive, except for the band 0a02 and CAMS NRT data: the SMAC product, whatever the source of ancillary data, tends to overestimate the TOC reflectance compared to the reference. The climatology is the source of ancillary data giving the less accurate results. CAMS NRT (Table 7 in Annex 3: Statistics of Surface Reflectances Comparison.) seems better than MERRA-2 (Table 8 in Annex 3: Statistics of Surface Reflectances Comparison.) for shorter wavelengths (<600nm) whereas MERRA-2 seems better for longer wavelengths (>700nm).

Precision (Figure 4) and uncertainty (Figure 5) are obviously strongly correlated. Precision and uncertainty values are typically around 0.025 (~ 4 times higher than the accuracy on average). At shorter wavelengths, their values are higher. Better results are obtained for CAMS and MERRA-2 data without significant differences (Table 7and Table 8 in Annex 3: Statistics of Surface Reflectances Comparison.).

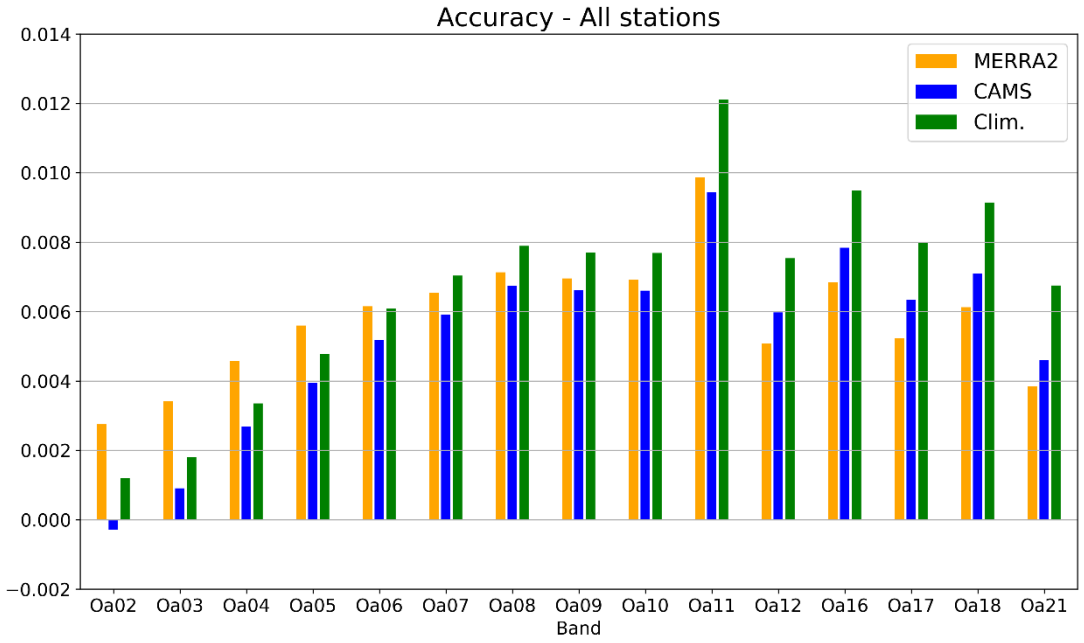


Figure 3: OLCI surface reflectances accuracy for different sources of ancillary data (MERRA2 reanalysis, CAMS NRT and a climatology). Only cloud free cases with an AOT lower than 1.0 and a SZA lower than 65° are kept.

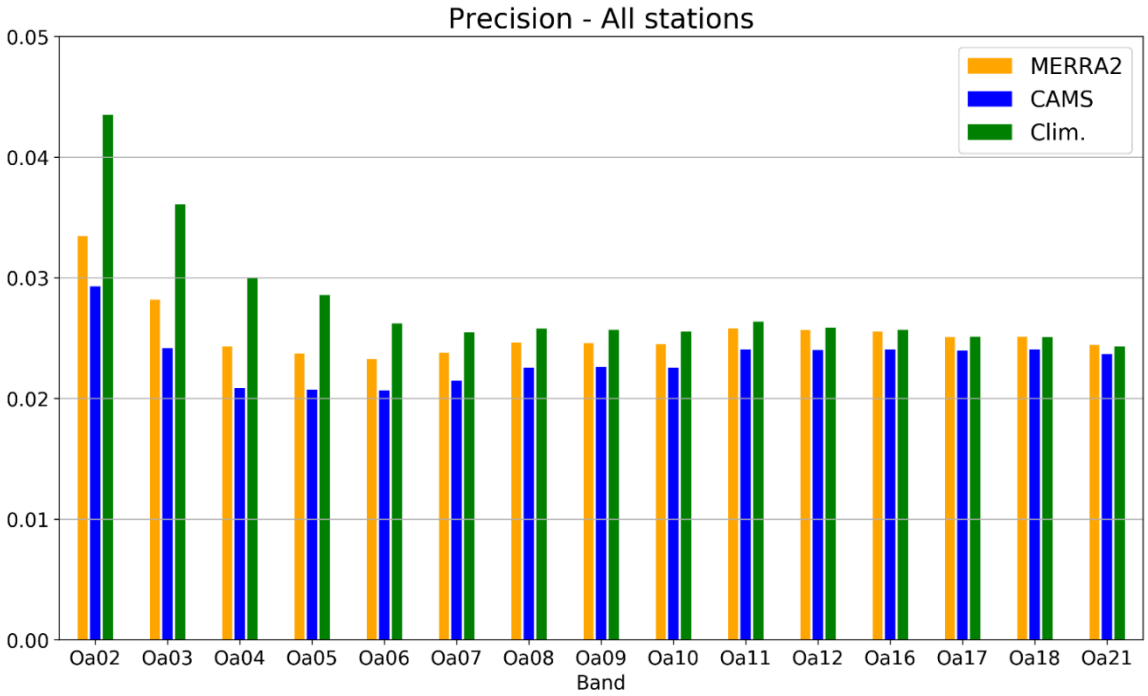


Figure 4: same as Figure 3 but for the OLCI surface reflectances Precision

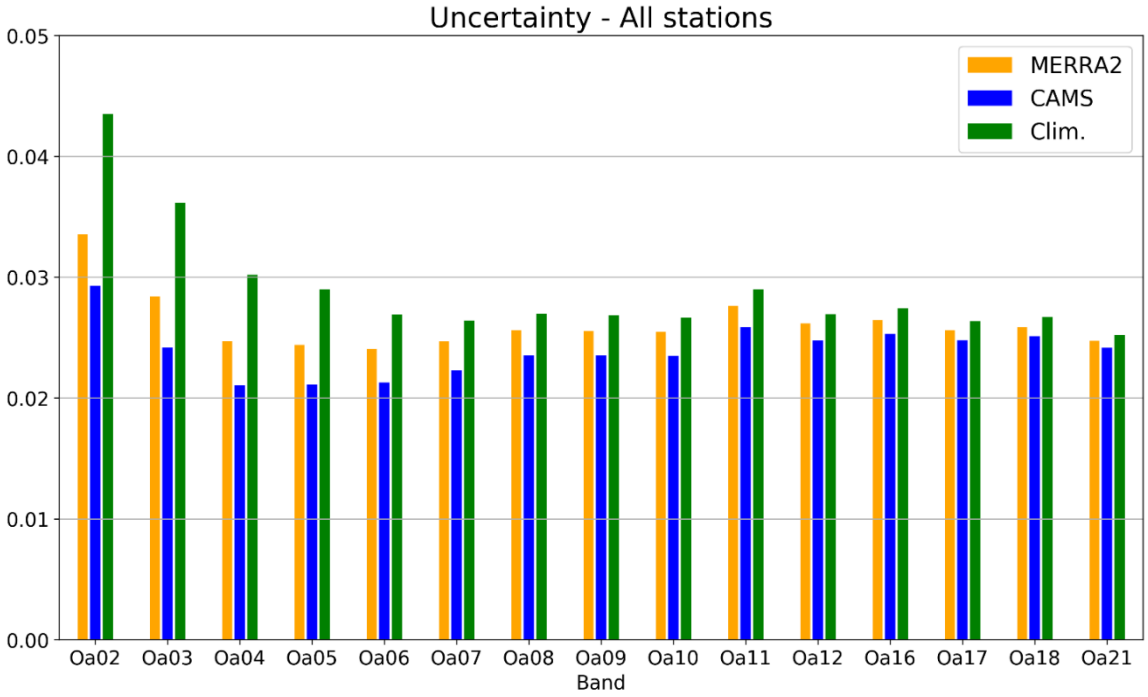


Figure 5: same as Figure 3 but for the OLCI surface reflectances Uncertainty

Figure 6, Figure 7 and Figure 8 show the scatter plots of the TOC reflectances from SMAC processor using ancillary data as a function of the TOC reflectances obtained from 6S and AERONET data for two OLCI bands: Oa03 centred at 443 nm with a large contribution of Rayeigh scattering and a weak contribution of the surface and 0a17, centred at 865 nm, with a poor contribution of Rayleigh scattering and a high signal from the surface. At 443 nm (band Oa03), the contribution of the atmosphere to the TOA signal is much more important than at 865 nm (Oa17). The absolute value of the uncertainty is approximately the same for the two bands but its relative value differs a lot because of the surface reflectance (35% for MERRA-2 and 30% for CAMS NRT for band Oa03, around 8% for MERRA-2 and CAMS NRT for band 0a17 in Table 7 and Table 8 in Annex 3: Statistics of Surface Reflectances Comparison.). The erroneous points for band Oa03, i. e. high values of SMAC TOC reflectances compared to low values of 6S reflectances, are due to large overestimation of the AOT by ancillary data compared to AERONET. In the opposite, low values of SMAC TOC reflectances compared to high values of 6S reflectances, are due to large underestimation of the AOT.

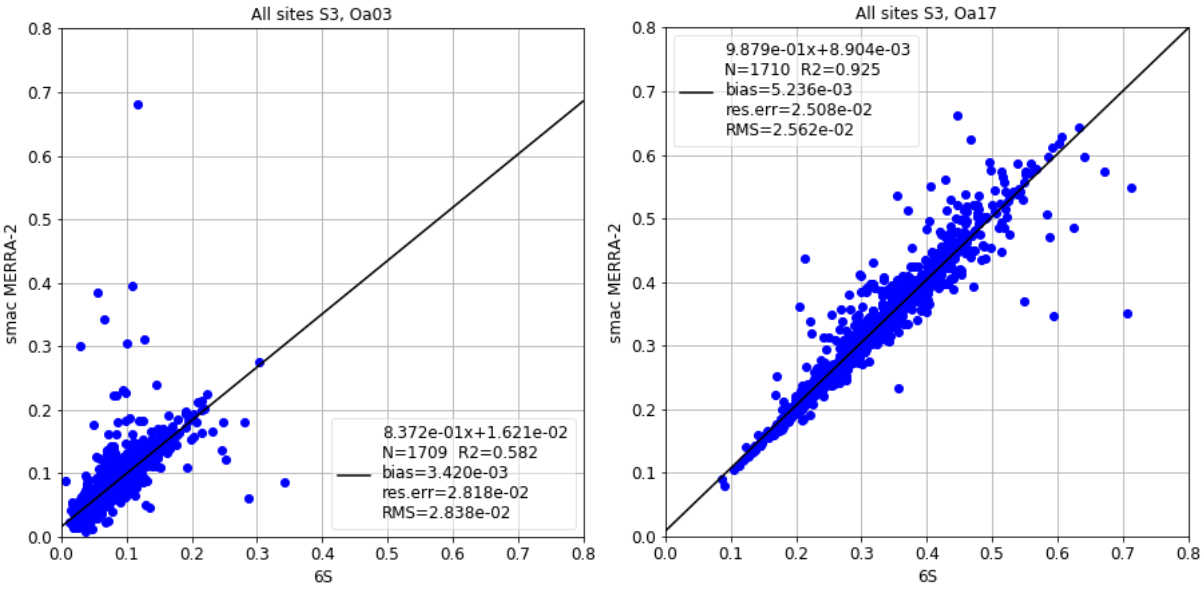


Figure 6: SMAC processor TOC reflectances using MERRA-2 versus TOC reflectances calculated with 6S and AERONET data. On the left, results are for the band Oaa03 (443 nm) and, on the right, for band 0a17 (865 nm).

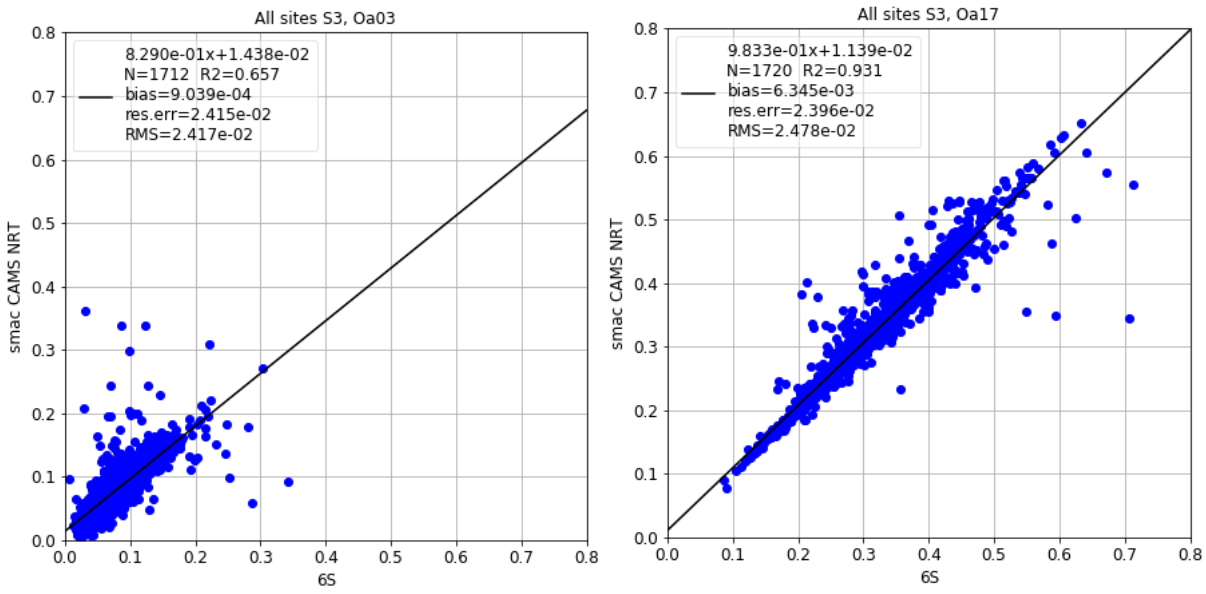


Figure 7: Same as Figure 6 for TOC reflectances using CAMS NRT.

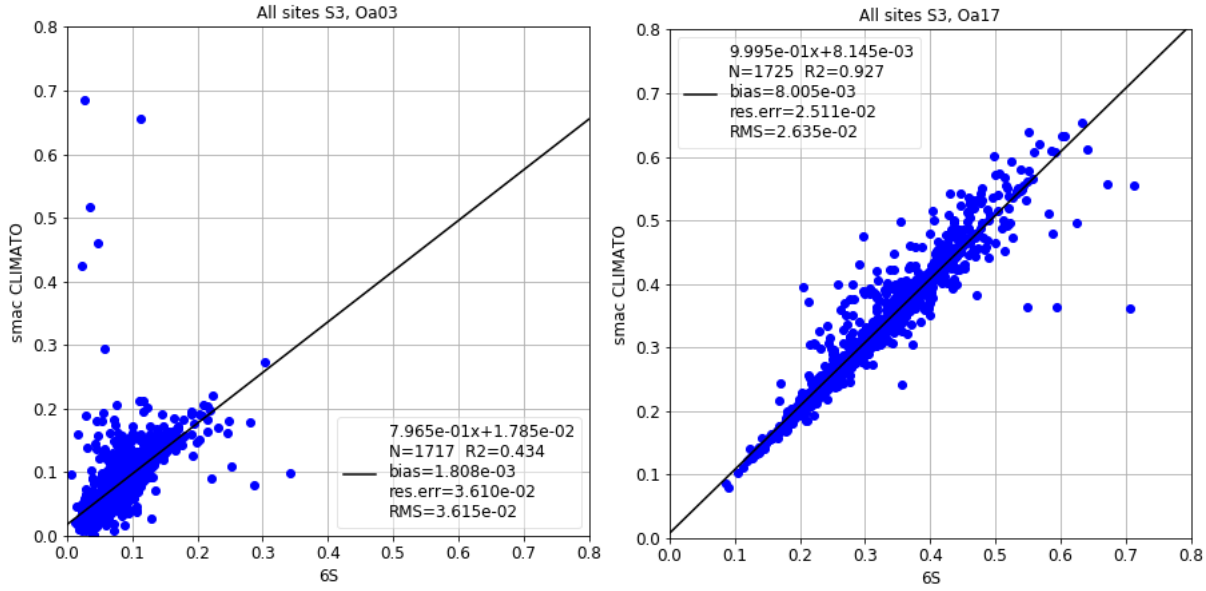


Figure 8: Same as Figure 6 for TOC reflectances using a climatology.

Figure 9 allows to check the variation of the accuracy over the different AERONET sites and locations. It appears clearly that the climatology is not performing well for very polluted and turbid sites as Beijing or Ilorin. MERRA2 and CAMS NRT gives similar results for almost all stations.

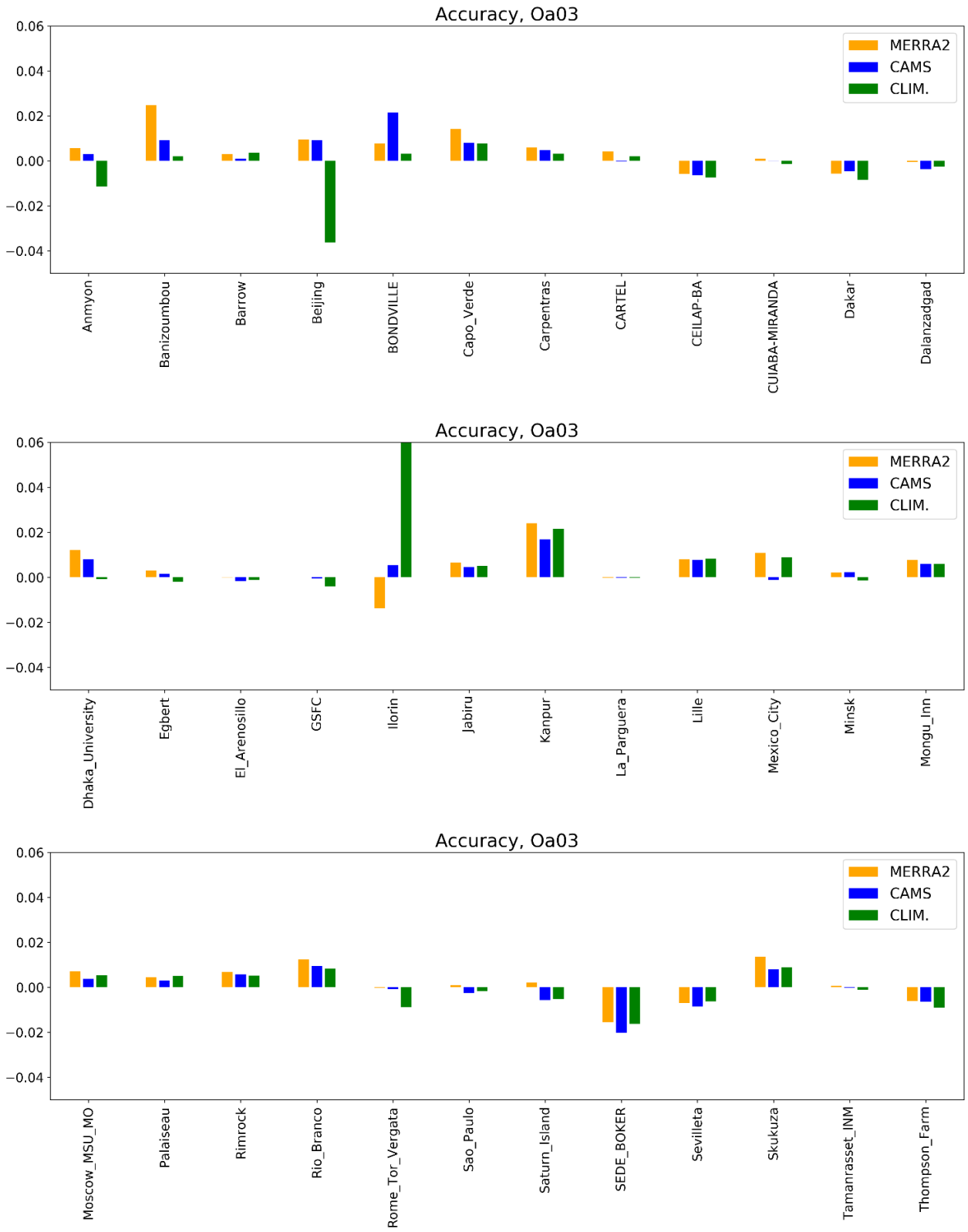


Figure 9: Value of the accuracy for the different AERRONET sites for the band Oa03

3.2.1.2 Statistics for all clear-sky cases

In this section, is investigated the effect of relaxing the filters on valid TOC reflectances in order to obtain more products. If no filters on AOT and solar zenith angles are applied, the impact on the accuracy, the precision and the uncertainty is shown in Figure 10, Figure 11 and Figure 12, respectively. Including all pixels, the number of match-ups only increases of about 100 cases. At the first glance, relaxing the filtering seems to be to the detriment of MERRA-2 ancillary data, whereas the impact is lower on accuracy obtained for CAMS NRT or climatology even if now climatology seems to be the better source with respect to precision and uncertainty. But looking more deeply at the datasets and their characteristics, we remark that, for climatology, no value of AOT greater than 1.0 is found whereas the number of MERRA-2 AOT larger than 1 is twice (21) than for CAMS NRT (11). These high values of AOT have a large impact on the quality of the product as shown in Figure 13. For AOT larger than 1.0, the accuracy is degraded by a factor 100 for short wavelengths and by a factor 20 for longer wavelengths (Figure 13). Precision and uncertainty are about 0.15 (Figure 14) and 0.25 (Figure 15), respectively, whereas they were about 0.03 for AOT smaller than 1 (Figure 11, Figure 12). These values clearly underline that the performance of SMAC are really degraded for high AOT.

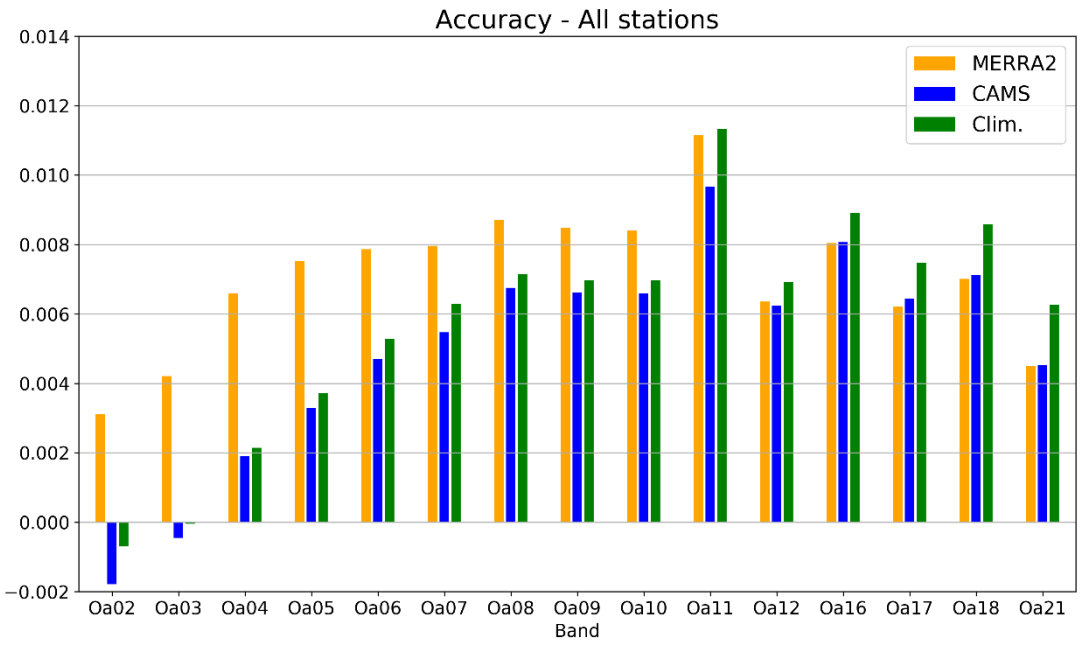


Figure 10: OLCI surface reflectances accuracy for different sources of ancillary data (MERRA2 reanalysis, CAMS NRT and a climatology). All cloud free pixels are kept (no filtering on AOT and SZA compared to Figure 3)

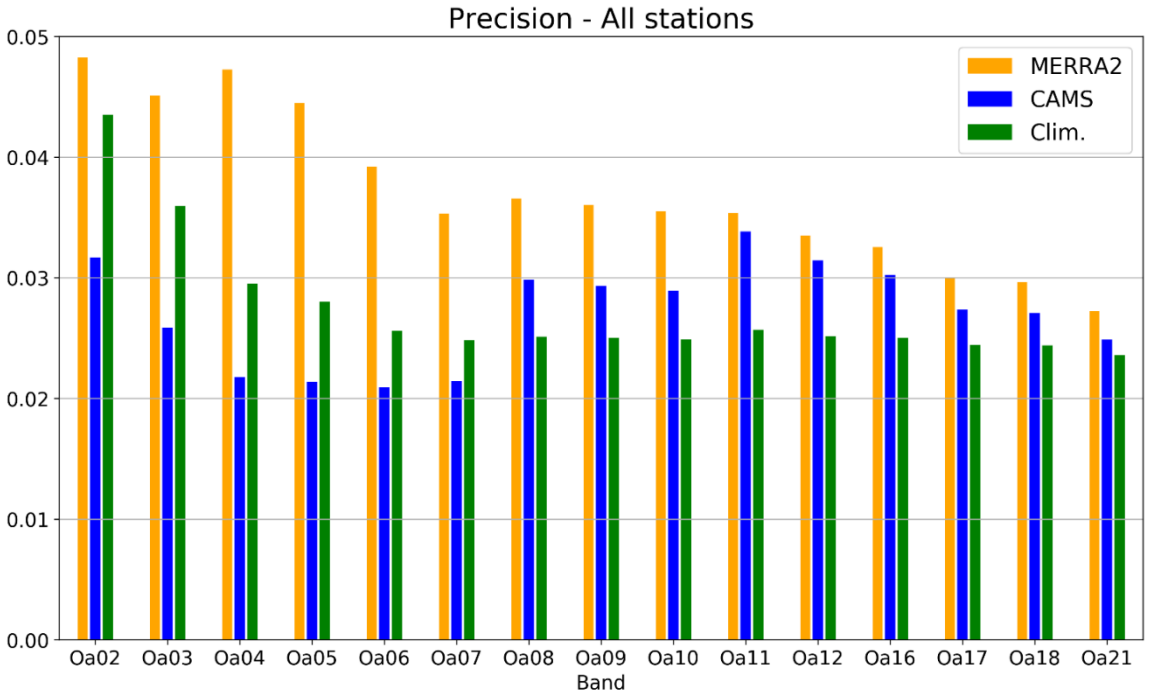


Figure 11: Same as Figure 10 but for the OLCI surface reflectances Precision. All cloud free pixels are kept (no filtering on AOT and SZA compared to Figure 4).

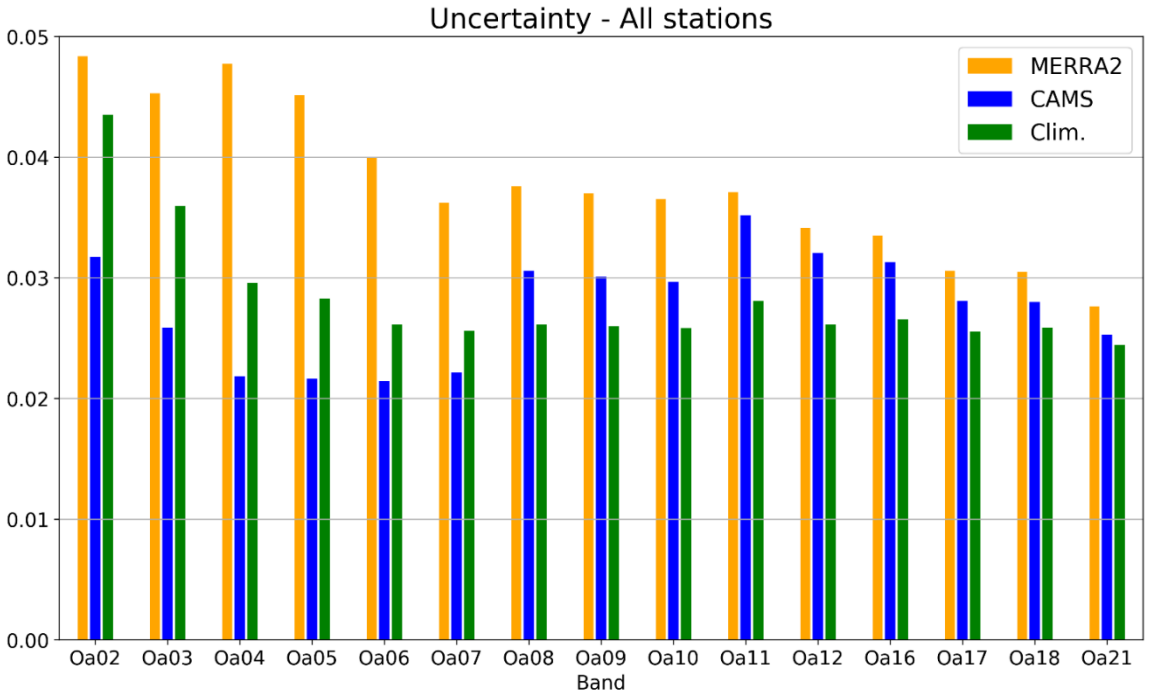


Figure 12: Same as Figure 10 but for the OLCI surface reflectances Uncertainty. All cloud free pixels are kept (no filtering on AOT and SZA compared to Figure 5).

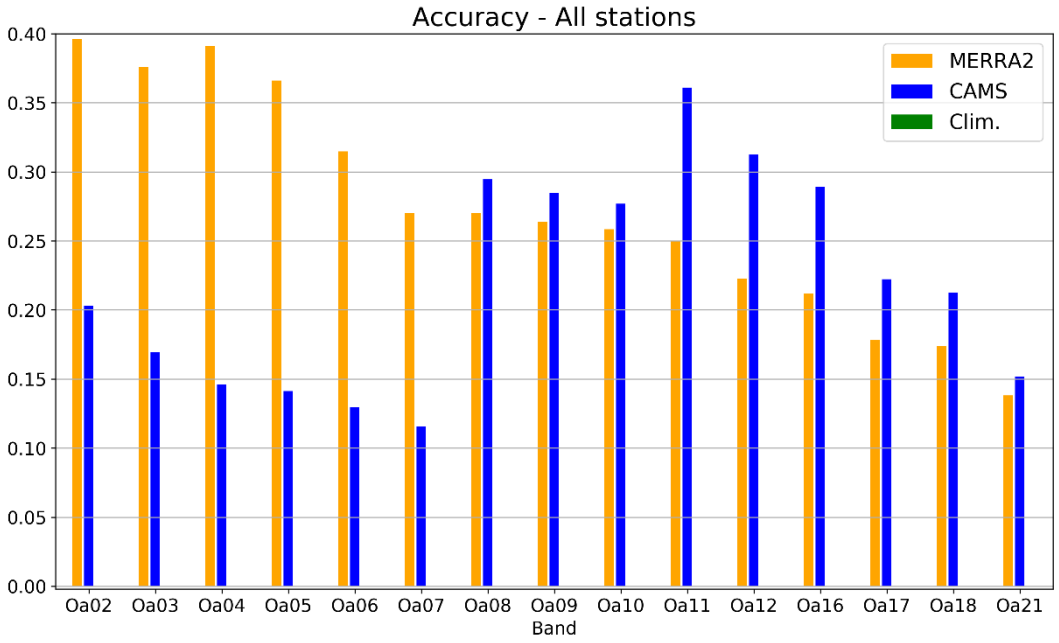


Figure 13: OLCI surface reflectances accuracy for different sources of ancillary data (MERRA2 reanalysis and CAMS NRT). Only cloud free cases with an AOT greater than 1.0 and SZA lower than 65° are kept. The results for the climatology are not shown because no AOT greater than 1.0 is found in the climatology

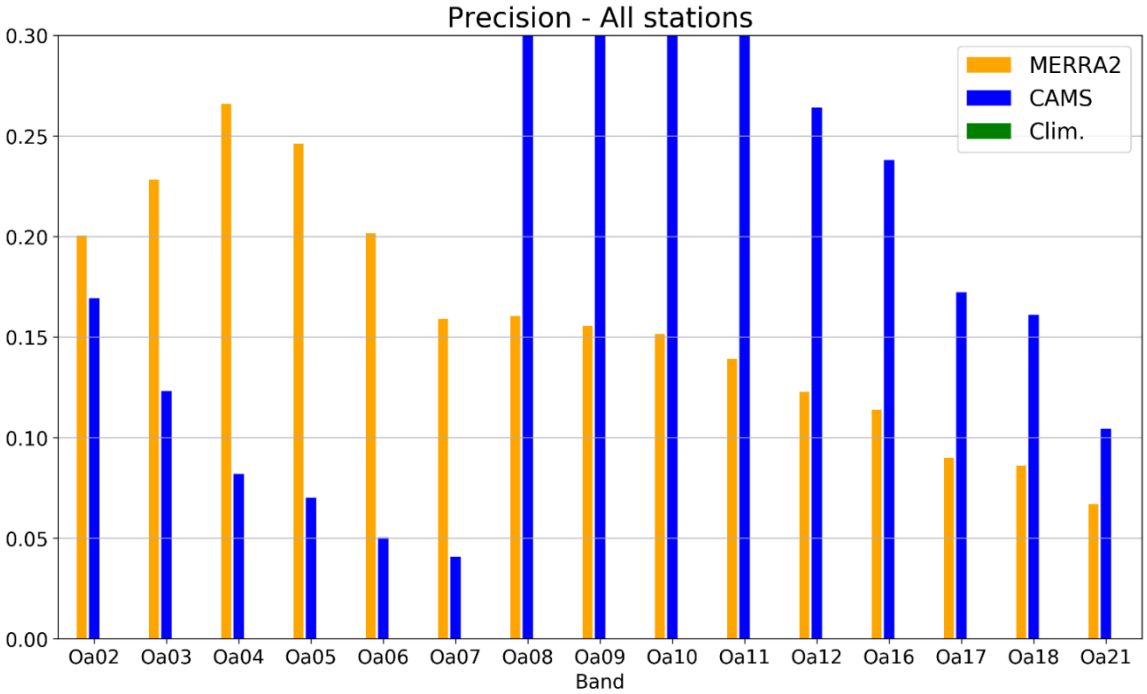


Figure 14: same as Figure 13 but for the OLCI surface reflectances Precision

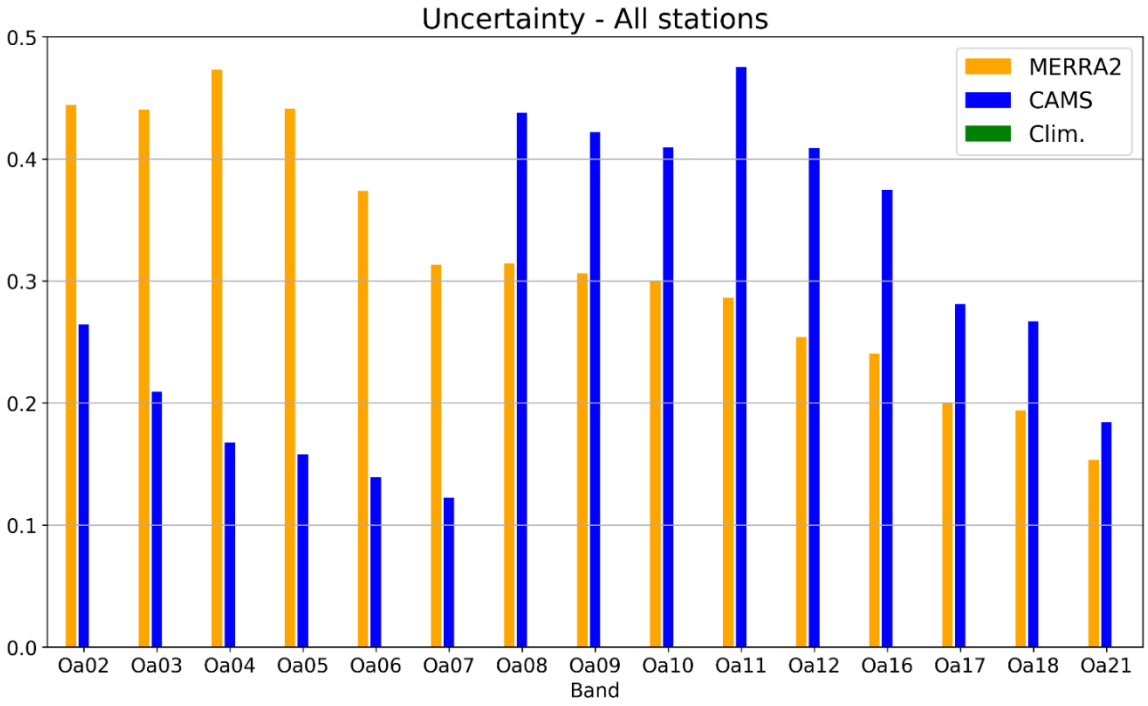


Figure 15: same as Figure 13 but for the OLCI surface reflectances Uncertainty.

If we focus on products calculated with large solar zenith angles (i. e. SZA > 65°), the number of match-ups is 120, for all AERONET sites and the whole year of interest. The weak number of match-ups can be explained by the lack of AERONET inversion products availability: the inversion of microphysics properties is only performed 5-6 times per day and, moreover, the inversion is not running for solar grazing angles (the limit value of SZA varies from a AERONET site to another, but the typical value is around 75°). This is illustrated in Figure 16 for the AERONET station “Lille”. SZA is larger than 75° between the 23 November (day 176) and 24 January (day 238). No 6S AERONET cases is available between day 168 (15 November) and day 183 (30 November) whereas retrieval of TOC reflectances from SMAC and MERRA-2 are available. We can also see that, for OLCI band Oa02, negative values of the TOC reflectance are obtained with MERRA-2 ingested into SMAC. Among the 120 match-ups with 6S AERONET, more than 70 negative TOC reflectances for SMAC with MERRA-2 were obtained for the band Oa02, and about 20 negative values for band Oa03(no negative TOC reflectance for other bands). This number of 70 negative values over 120 cases (i. e. 58%) must be compared to the number of 2 negative values over 1700 cases (i.e. 0.11%) when filtering the data on AOT (less than 1.) and SZA (less than 65 degrees). Note that the lack of TOC reflectance between day 183 and 256 over Lille station (Figure 16) is due to cloud contamination.

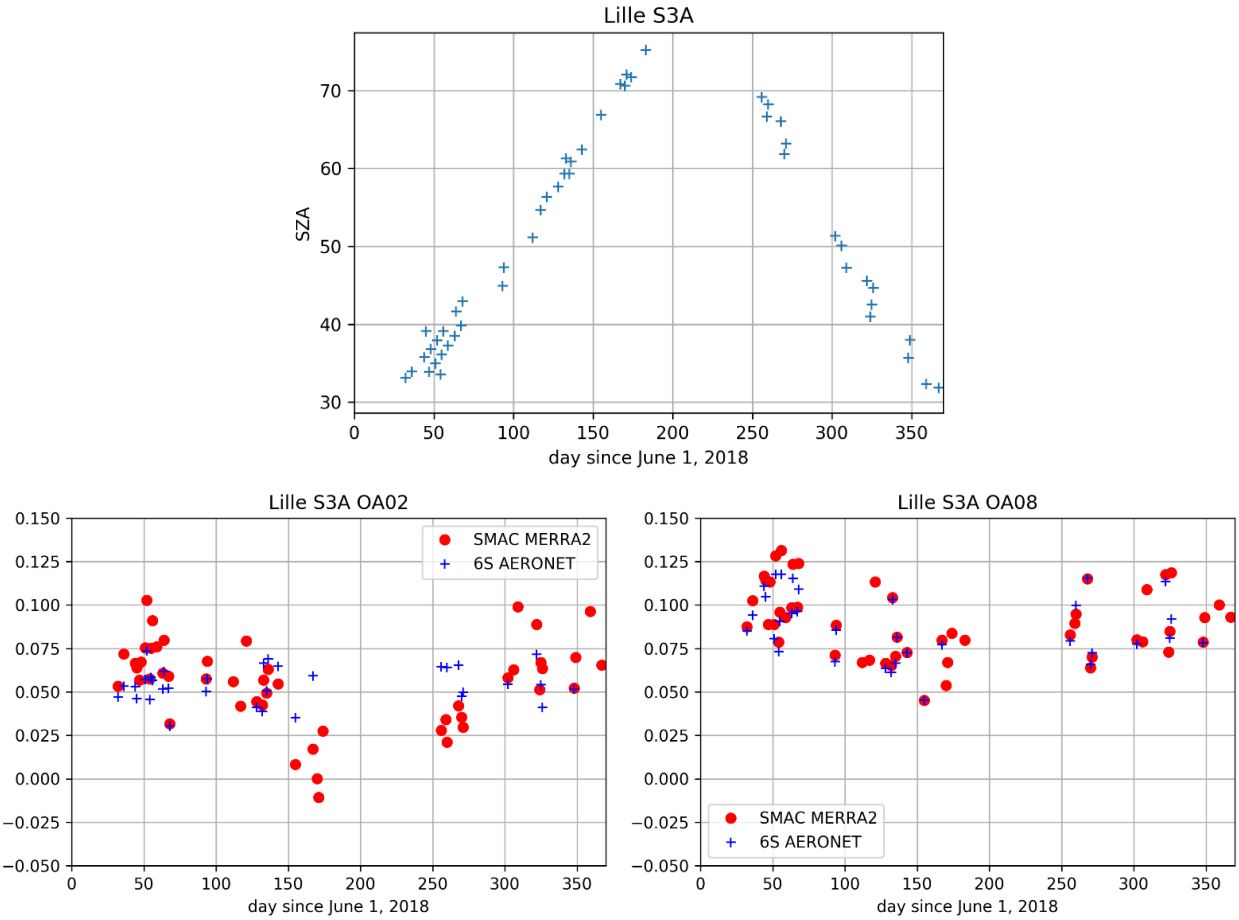


Figure 16: Times series of SZA (top panel), TOC reflectance for band Oa02 (bottom left) and TOC reflectance for band Oa08 (bottom right) for cloud-free pixel over AERONET station located in Lille.

Excluding negatives values gives statistical results on accuracy (Figure 17), precision (Figure 18) and uncertainty (Figure 19) with completely different trends than those obtained previously. Accuracy (mean bias) is now negative for shorter wavelengths (bands Oa02 to Oa08) and around zero for the other wavelengths (Figure 17). At shorter wavelengths, absolute value of the accuracy is almost 10 times higher than values obtained for SZA less than 65° (Figure 3). For all bands, the precision is better than the value for SZA less than 65° and the uncertainty is mostly driven by the systematic bias at shorter wavelengths (accuracy) and by the precision at longer wavelengths.

Results obtained for large solar zenith angles have to be handled with care because of the complexity of the radiative transfer calculations for high airmass and grazing angles:

- The plane – parallel assumption used in 6S is not valid anymore
- From the top of atmosphere to the surface, we make the assumption that the horizontal distribution of atmospheric parameters is constant despite of very long distance.

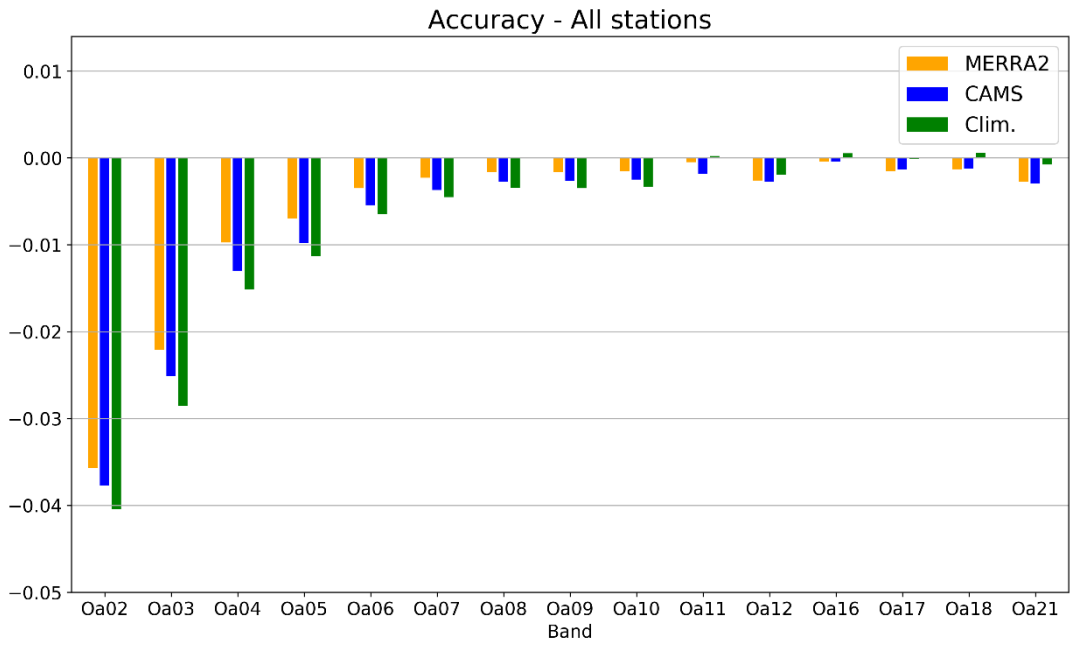


Figure 17: OLCI surface reflectances accuracy for different sources of ancillary data (MERRA2 reanalysis, CAMS NRT and climatology). Only cloud free cases with SZA larger than 65° are kept.

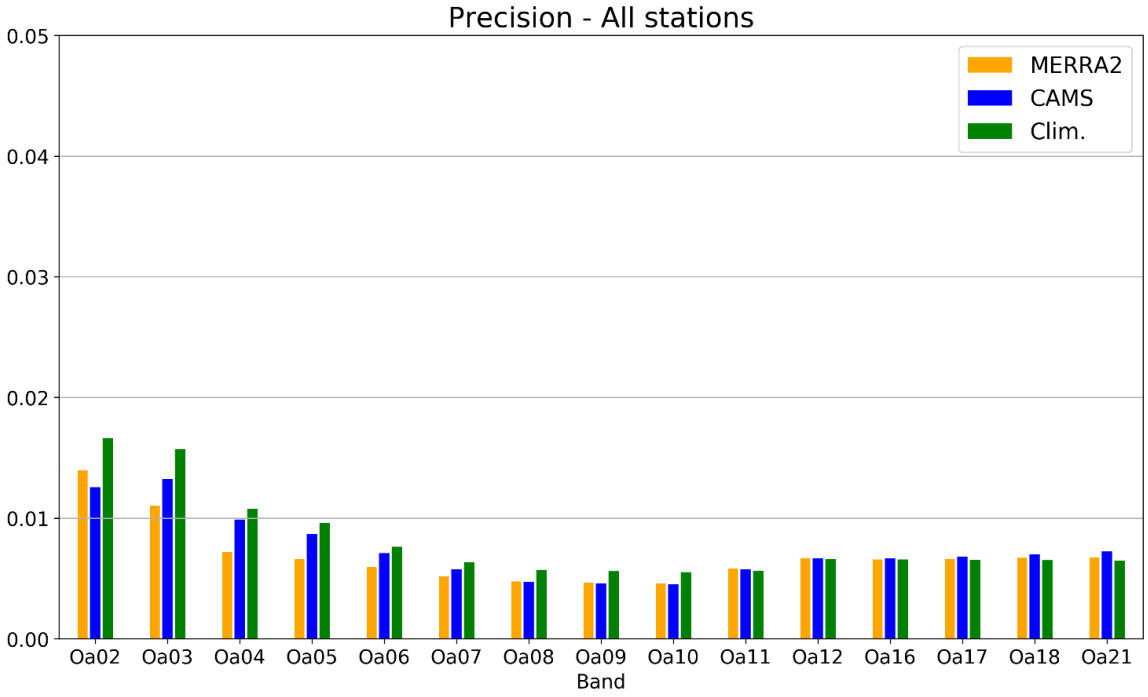


Figure 18: Same as Figure 17or the OLCI surface reflectances Precision

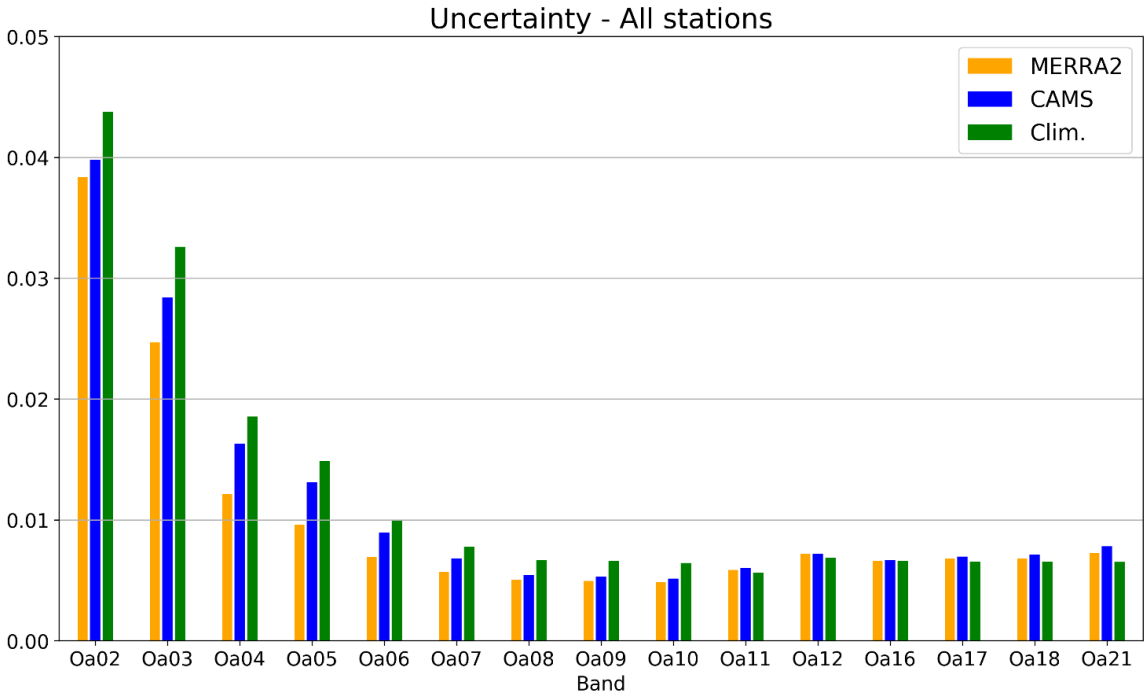


Figure 19: Same as Figure 17 for the OLCI surface reflectances Uncertainty.

3.2.2 Surface reflectance from SLSTR instrument

Among the five bands of SLSTR atmospherically corrected and used for the generation of bio-physical products, three are similar to spectral bands of OLCI: band S1 (centred at 554 nm) is very close to the band Oa06, Band S2 (660 nm) is spectrally similar to band Oa08 as well as band S3 (868 nm) with band Oa17. The two other bands, S5 (1613 nm) and S6 (2255 nm) are in the shortwave infrared not covered by OLCI.

For the OLCI instrument (see Section 3.2.1), we have seen that using CAMS NRT or MERRA-2 atmospheric parameters as inputs of the AC processor SMAC give similar results. The intercomparison with 6S and AERONET inversion products have shown that the accuracy, the precision and uncertainty are very closed each other when filtering data with high AOT. For this reason, we only show, for SLSTR sensor, results for the best case: only pixels with an AOT smaller than 1.0 and SZA smaller than 65° are kept.

Results of the validation of the atmospheric corrections of SLSTR bands in terms of accuracy precision and uncertainty are shown in Figure 20, Figure 21 and Figure 22, respectively. Not surprisingly results for band S1, S2 and S3 are very close to the results obtained for corresponding OLCI bands (Figure 3 to Figure 5). Using CAMS NRT or MERRA-2 reanalysis, accuracy is better than 0.006 for

band S5 and better than 0.004 for band S6. Precision and uncertainties are in the order of 0.017 and 0.012 for band S5 and S6, respectively. For the precision and the uncertainty, there is no significant difference between the three sources of ancillary data for the shortwave infrared bands S5 and S6.

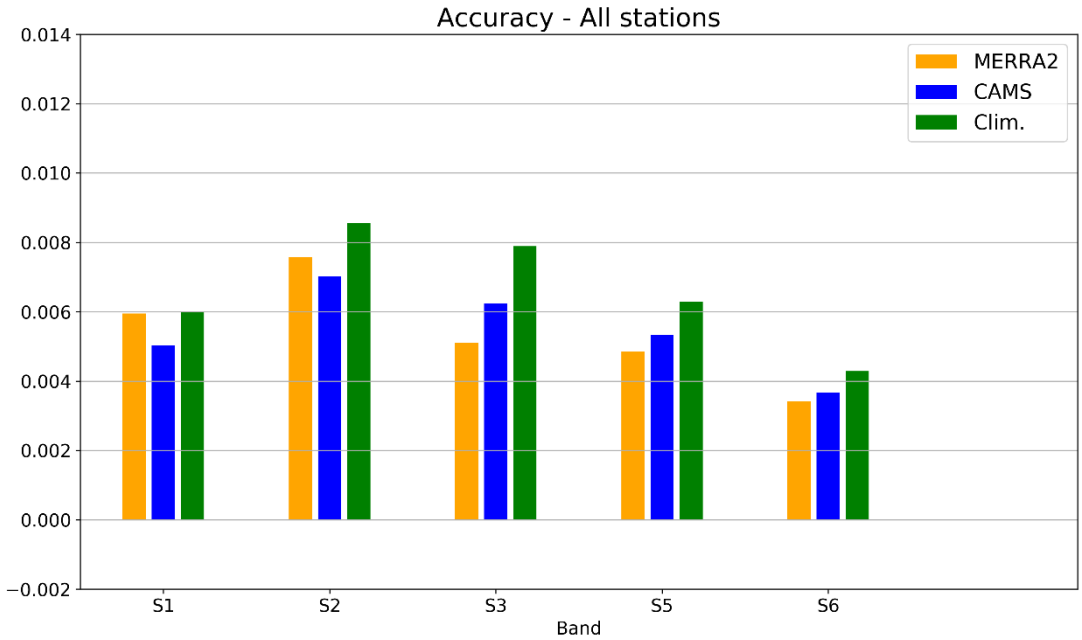


Figure 20: SLSTR surface reflectances accuracy for different sources of ancillary data (MERRA2, CAMS NRT and a climatology). Only cloud free cases with an AOT less than 1.0 and SZA less than 65° are kept

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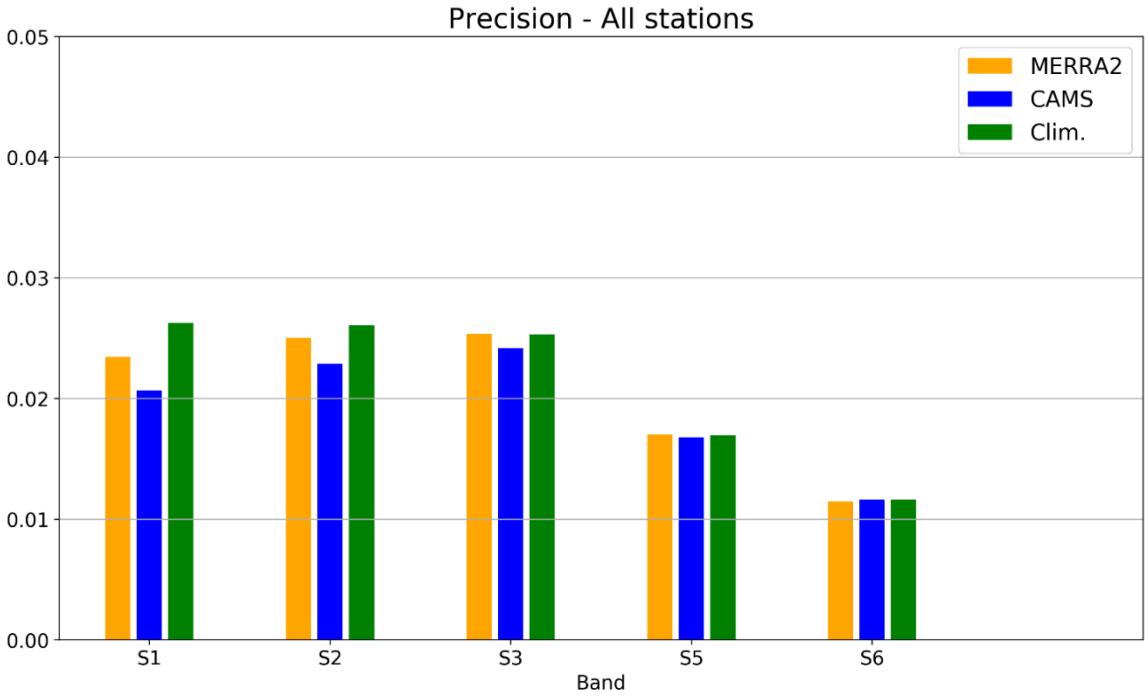


Figure 21: Same as Figure 20 for SLSTR surface reflectances precision.

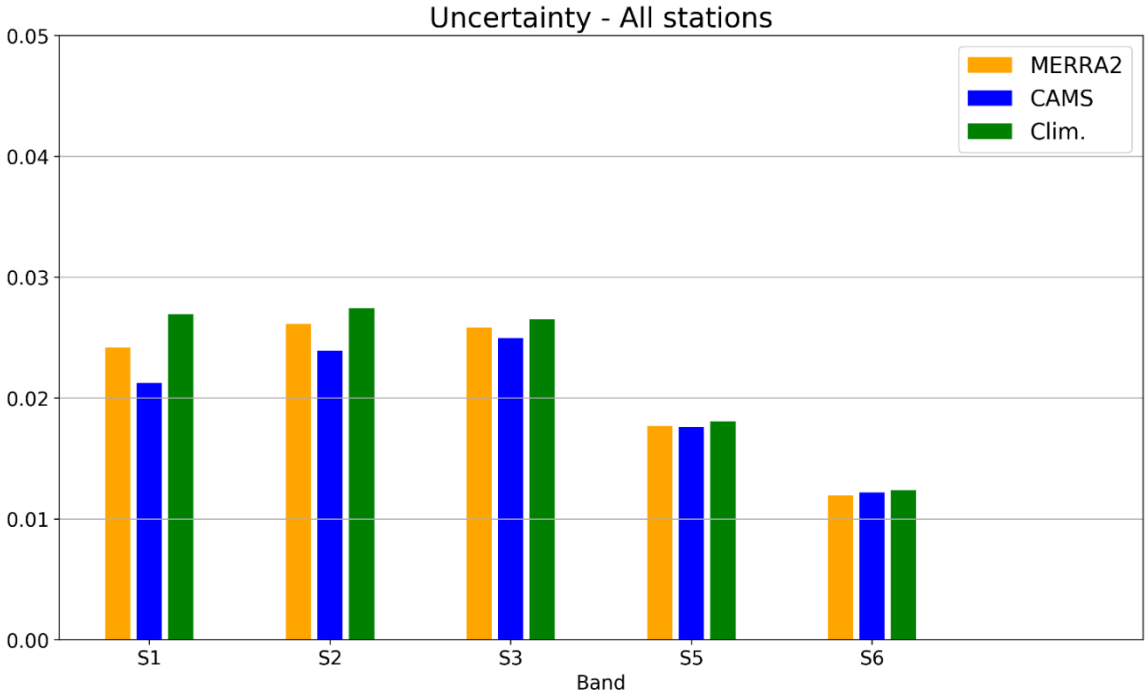


Figure 22: Same as Figure 20 for SLSTR surface reflectances uncertainty.

3.2.3 Discussion on the aerosol model

A unique continental aerosol model has been used to calculate and infer the TOC reflectances with the SMAC processor. Part of the discrepancy between SMAC outputs and the reference (6S using AERONET data) should come from the choice of this model because the type of aerosol can vary spatially and temporally due to wind (desert dust storm), sources of anthropogenic aerosols (forest fire, pollution), natural events (volcanic ashes) and others.

Radiative properties of aerosol can differ a lot from a model to another, in particular:

- The angular distribution of the radiation scattered by aerosol, i.e., the phase function
- The absorption.

Absorption of the aerosol is usually quantified by the single scattering albedo (SSA) that gives the ratio of absorbed radiation at each interaction between radiation and a particle and its value is between 0 And 1. If SSA is 1, no radiation is absorbed. Absorption increases when SSA decreases.

The value of the SSA of the continental aerosol model can differ significantly from the value measured by AERONET as illustrated in Figure 23: for Carpentras, the continental model tends to overestimates the aerosol absorption; for Banizoumbou, the SSA of the continental model can be, on average, representative at 440 nm but it is underestimated at 865 nm.

The uncertainty due to the aerosol model can be quantified by performing for example a huge amount of Monte Carlo radiative transfer simulation with different models. However, such analysis was beyond the scope of the present analysis. The amplitude of this uncertainty will vary a lot (i) from a location to another and (ii) temporally. Moreover, in an operational near real time system, it won't be possible to take into account extreme and exceptional aerosol events (volcanic ash, industrial accident pollution, ...).

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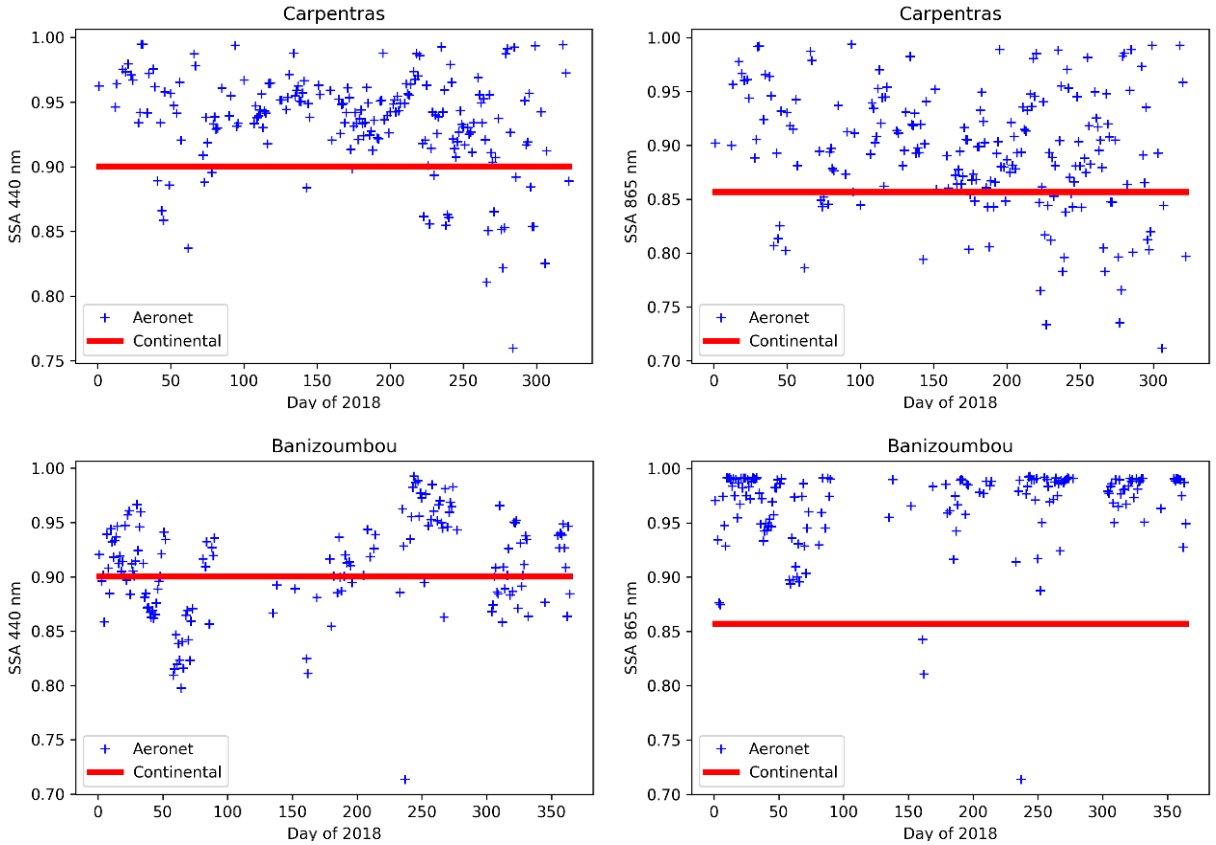


Figure 23: Time series of AERONET daily single scattering albedo (SSA) at 440 nm (left) and at 865 nm (right) for Carpentras (France) (top) and Banizoumbou (Niger) (Bottom). The value of the SSA of the continental model is indicated by the red line.

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4 PERFORMANCES OF THE CGLOPS-1 TOP OF CANOPY REFLECTANCES PRODUCTS

4.1 TOP OF CANOPY REFLECTANCES PRODUCTS

Two versions of the CGLOPS-1 TOC-r product, version 1 (V1) and version 2.0 (V2), are under consideration. As detailed in Section 2.2.2, the only difference in the atmospheric corrections scheme between V1 and V2 concerns the TOC reflectances error bars because, as of version 2.0, the error due to the uncertainty on the aerosols model is added to the error bars [CGLOPS1_ATBD_S3-AC-V1.1].

However, other changes are applied to retrieve successive versions of TOC reflectances:

1. From version 2.0 upwards, correction factors of calibration are applied to SLSTR TOA reflectances: 0.97, 0.98; 0.99, 1.11 and 1.13 for, respectively, bands S1, S2, S3, S5 and S6, as recommended by ESA [S3MPC.RAL.TN.020]
2. From version 2.3 upwards,
    a. a correction factor of calibration equal to 0.98 is applied on all Sentinel-3A OLCI bands, except band Oa21, to reduce the brightness of OLCI-A acquisitions by 2% and make them more consistent with the OLCI-B data. This calibration adjustment results from analysis of OLCI data acquired during the tandem phase (Clerc et al., 2020).
    b. Radiometric uncertainties on OLCI TOA radiances are included into the Level 1B input files instead of taken a relative value of 2% for previous versions.

The application of calibration adjustment factors has no impact on the AC scheme.

The change on TOA radiometric uncertainties does not affect the TOC reflectances but slightly the error bars on the TOC reflectances.

Consequently, the results found for TOC reflectance V2 and reported in the following section, i.e. APU metrics, are valid for TOC reflectance version 2.3. The errors bars on OLCI TOC reflectances version 2.3 would be slightly different than errors bars on OLCI TOC reflectances V2 but without prejudice to the conclusions.

Version 1 was the NRT operational version until 28th February 2025. The version 2.3 was deployed in NRT operations on 1st March 2025.

4.2 ACCURACY, PRECISION AND UNCERTAINTY

In the previous chapter, we have studied the impact of the source of ancillary data on the performance of the AC processing scheme. In this chapter, we present the results for the CGLOPS-1 TOC reflectance that are calculated by the CGLOPS-1 processing line [CGLOPS1_ATBD_S3-AC-V1.1] using:

- AOT and total column of ozone from CAMS NRT
- Surface pressure and total column of vapour from S3 L1B input files (source ECMWF) because they are only available with a delay of +3 days in CAMS NRT products.

The dataset for the version 1 is composed of 1 year of Sentinel-3A observations (July 2018 to June 2019) and 6 month (January to June 2019) of Sentinel-3B observations. For the version 2.0, the dataset is composed of 1 year (2020) of Sentinel-3A and Sentinel-3B observations. The period of the two versions is not the same but one year is enough to cover natural and seasonal variations. Moreover, to make more robust the comparison between the two versions, it has been decided to only keep the 41 AERONET stations that (i) have measurements for the two periods and (ii) with more than 10 days of clear sky observations. The size of the sample between V1 and V2 differs because for the V1 version only the pixel corresponding to the AERONET site is kept whereas a block of 3x3 pixels around the site is kept for V2.

Therefore, CGLOPS-1 V1 and V2 TOC reflectance products are very closed to TOC reflectances obtained with all atmospheric parameters from CAMS NRT. Accuracy, precision and uncertainty values found for OLCI and SLSTR, as shown in Figure 24 to Figure 29, are very closed to values obtained with CAMS NRT (Figure 3 to Figure 5 and Figure 20 to Figure 22). That is not surprising because only surface pressure and total column of water vapour are coming from another source of ancillary data.

Better accuracy is obtained for the version 1 for all bands, whatever OLCI or SLSTR sensor, whereas better precision and uncertainty are only obtained for smaller wavelengths. The difference is probably due to the fact that the period of analysis is not the same for the two versions. Inter-annual variation of the state of the atmosphere can explain the difference between the two sets of data used for V1 (July 2019 to June 2019) and V2 (all year 2020). The frequency of dust events and biomass burning can vary a lot from one year to another for example.

For SLSTR bands in the SWIR, precision and uncertainty are better for V2. It can be explained by the fact that coefficients of calibration adjustment were applied to TOA reflectances in V2 (1.11 for band S5 and 1.13 for band B6). Such high values increase the dynamic of the TOA signal and thus decrease the contribution of the atmospheric signal.

Globally, ACIX metrics calculated for the two versions are comparable and the difference is rather small compared to the absolute value: it is in particular true for the uncertainty metric.

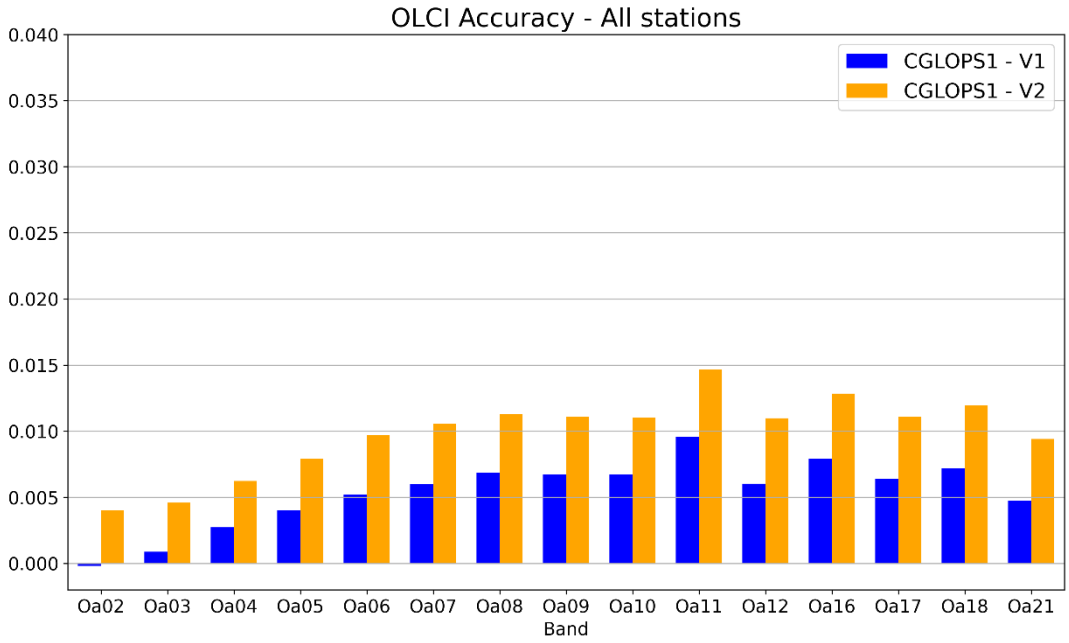


Figure 24: OLCI surface reflectances accuracy for the CGLOPS-1 products version 1 (in blue) and version 2 (in orange). Only cloud free cases with AOT less than 1.0 and SZA less than 65° are kept.

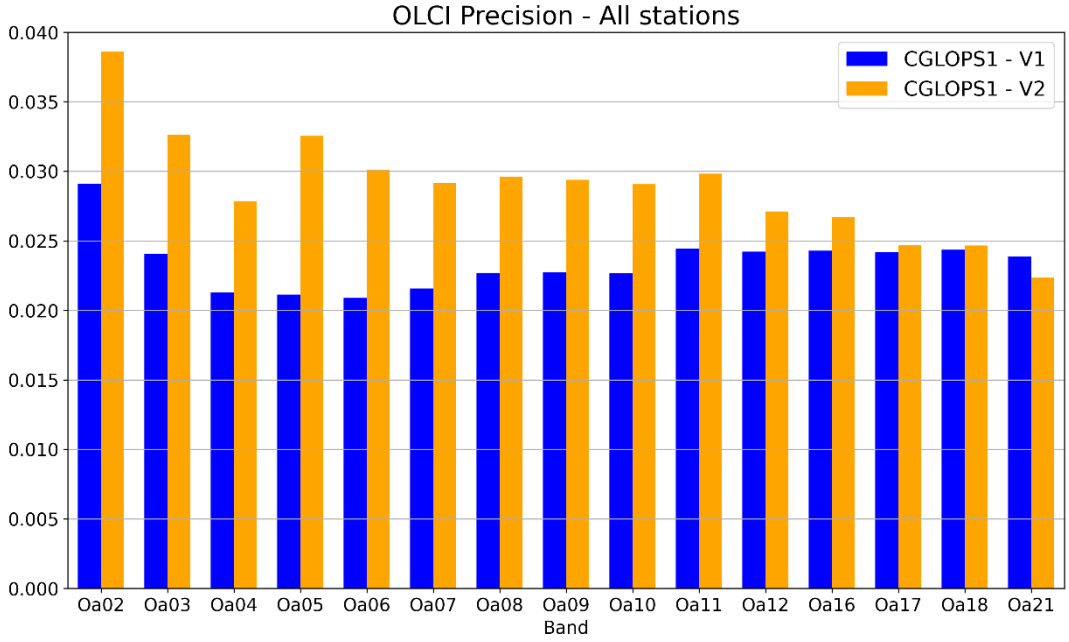


Figure 25: same as Figure 24 but for the precision

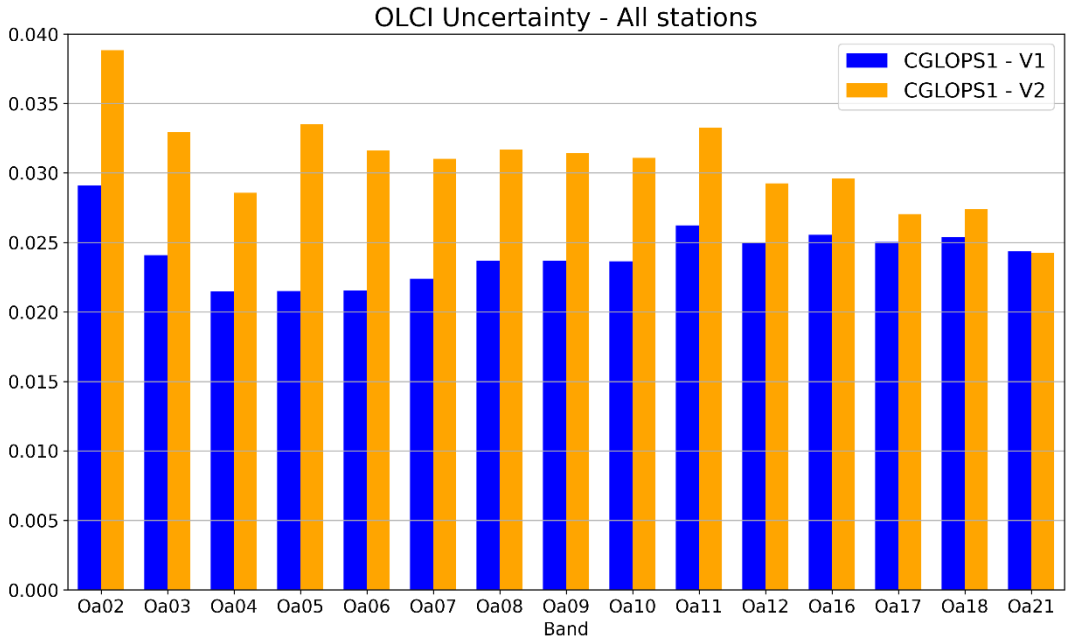


Figure 26: same as Figure 24 but for the uncertainty

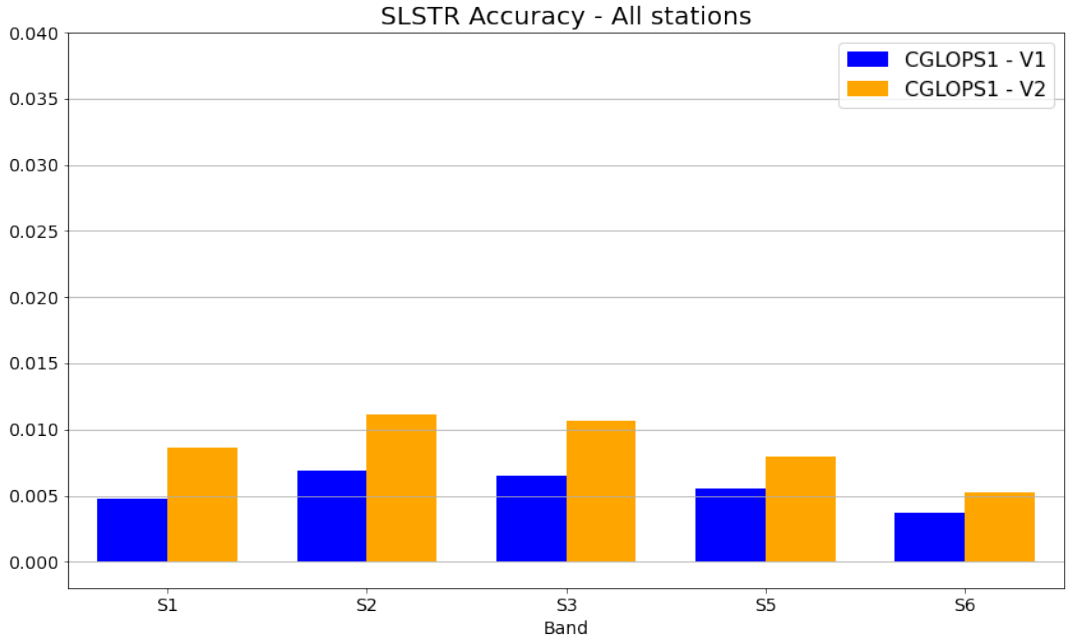


Figure 27: SLSTR surface reflectances accuracy for the CGLOPS-1 products (blue for version 1 and orange for version 2). Only cloud free cases with AOT less than 1.0 and SZA less than 65° are kept.

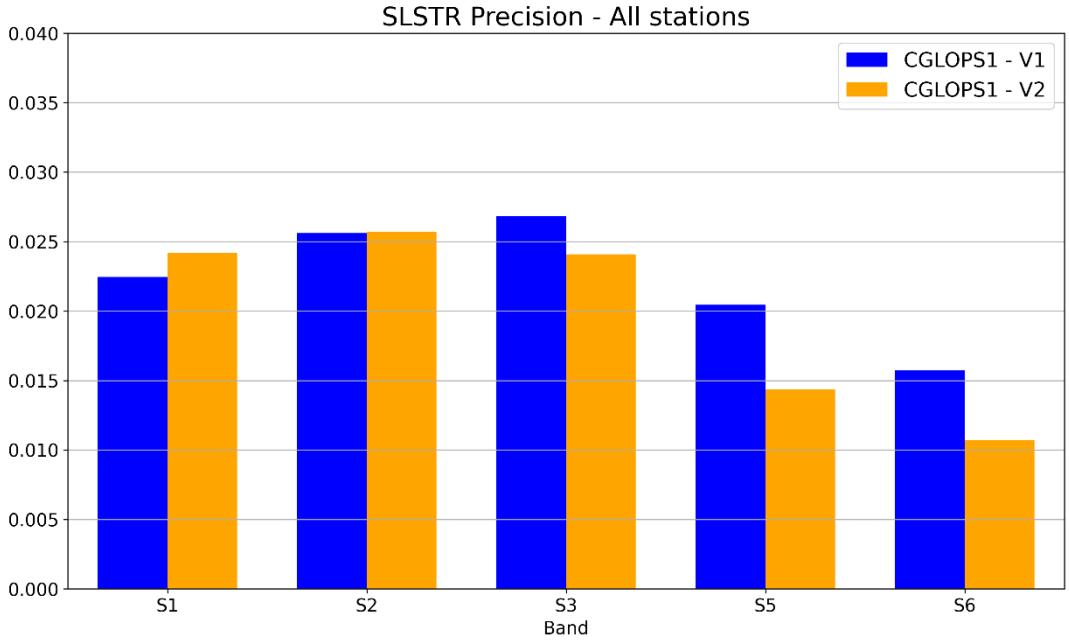


Figure 28: same as Figure 27 but for the precision

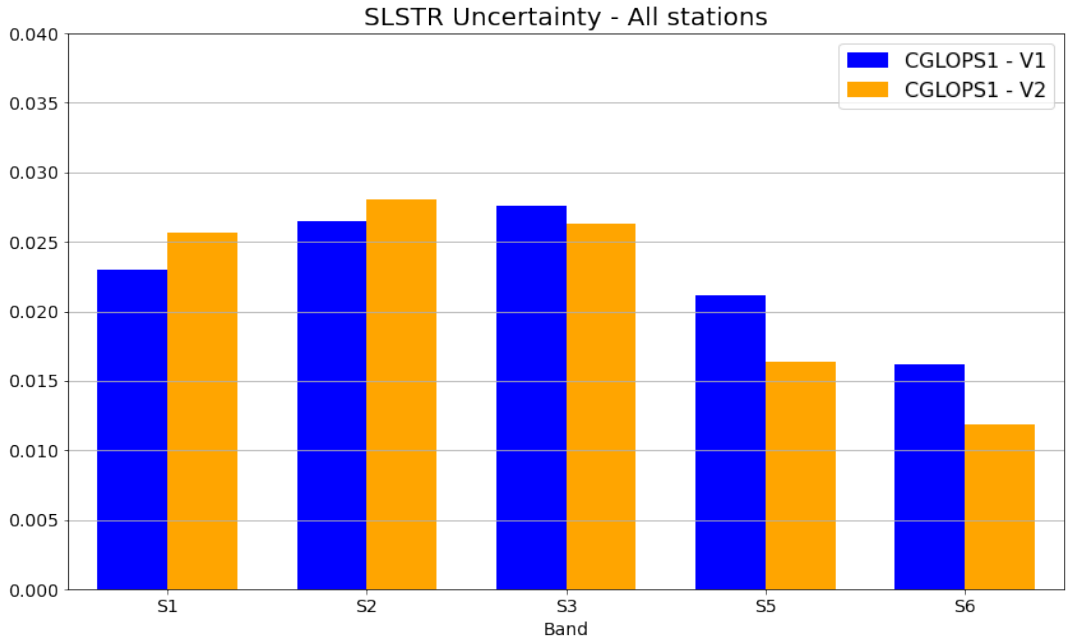


Figure 29: same as Figure 27 but for the uncertainty

These results are summarized in Table 3. Relative values (in %) and the correlation coefficient of the least square regression are also given. The mean bias in relative value is less than 5% for the version

1 and less than 7.5% for version 2 for all the bands. Relative values of precision and accuracy is higher than 20 % for short wavelength (Band 0a02, Oa03, Oa04 and Oa05) less than 510 nm. Their values decrease with the wavelengths to reach a minimum of 7% for version 1 (8% for version 2). By the same way, the correlation coefficient of the least square regression increases with the wavelength.

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Table 3: Statistics of the comparison between CGLOPS-1 version 1 (white line) and version 2 (green line) TOC reflectances and 6S using AERONET inversion products for each band of OLCI and SLSTR. N is the number of match-ups. A is the accuracy (and its relative value in %), P is the precision (and its relative value in %), U the uncertainty (and its relative value in %) and R^2 is the correlation factor. Only cloud free pixels, with a SZA smaller than 65° and AOT smaller than 1.0 are kept.

Band	N	A	A (%)	P	P(%)	U	U (%)	R^2
Oa02	1686	-0.000182	-0.27	0.029371	44.22	0.029371	44.22	0.400949
	21370	0.004025	5.83	0.038610	55.95	0.038819	56.25	0.330118
Oa03	1693	0.000884	1.11	0.024218	30.29	0.024244	30.32	0.654779
	21581	0.004590	5.56	0.032632	39.50	0.032953	39.89	0.645297
Oa04	1696	0.002732	2.92	0.020896	22.26	0.02109	22.46	0.812470
	21632	0.006218	6.41	0.027869	28.74	0.028555	29.45	0.721508
Oa05	1699	0.004001	3.92	0.020757	20.24	0.02116	20.63	0.843064
	21639	0.007921	7.46	0.032542	30.65	0.033492	31.55	0.695872
Oa06	1701	0.005203	3.91	0.020662	15.42	0.021336	15.92	0.904494
	21662	0.009698	7.01	0.030081	21.74	0.031605	22.84	0.830352
Oa07	1701	0.005971	4.01	0.021493	14.33	0.022326	14.88	0.944655
	21662	0.010544	6.73	0.029169	18.61	0.031016	19.79	0.913789
Oa08	1701	0.006859	4.40	0.022606	14.38	0.023627	15.03	0.950976
	21662	0.011292	6.85	0.029589	17.96	0.031670	19.22	0.927611
Oa09	1701	0.006733	4.29	0.022641	14.30	0.023623	14.92	0.952596
	21662	0.011086	6.68	0.029379	17.69	0.031402	18.91	0.931025
Oa10	1701	0.006721	4.24	0.022604	14.13	0.023582	14.74	0.953578
	21662	0.011017	6.57	0.029067	17.33	0.031085	18.53	0.933657
Oa11	1701	0.009581	4.85	0.024155	12.13	0.025984	13.05	0.934270
	21662	0.01466	7.16	0.029834	14.57	0.033244	16.24	0.916866
Oa12	1701	0.006028	2.19	0.024053	8.67	0.024821	8.94	0.92462
	21662	0.010989	3.90	0.027099	9.63	0.029242	10.39	0.922118
Oa16	1701	0.007896	2.77	0.024108	8.39	0.025397	8.83	0.926637
	21662	0.012825	4.40	0.026679	9.14	0.029601	10.14	0.927662
Oa17	1701	0.006418	2.09	0.023992	7.76	0.024855	8.04	0.930743
	21662	0.011095	3.54	0.024670	7.87	0.027050	8.63	0.939468
Oa18	1701	0.007192	2.31	0.024118	7.70	0.025184	8.04	0.930705
	21662	0.011959	3.77	0.024657	7.77	0.027405	8.64	0.940089
Oa21	1701	0.004742	1.42	0.023731	7.04	0.024202	7.18	0.940469
	21662	0.009393	2.75	0.022367	6.54	0.024259	7.09	0.955539
S1	1661	0.004783	3.49	0.022244	16.08	0.022780	16.47	0.894103
	21588	0.008608	6.34	0.024167	17.79	0.025655	18.88	0.882430
S2	1661	0.006869	4.29	0.025591	15.79	0.026502	16.35	0.939463
	21586	0.011101	6.77	0.025726	15.68	0.028619	17.08	0.944985

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S3	1661	0.006482	2.12	0.026708	8.63	0.027484	8.88	0.924772
	21605	0.01648	3.46	0.024081	7.81	0.026330	8.54	0.950884
S5	1643	0.005577	2.22	0.020545	8.03	0.021252	8.30	0.963113
	21341	0.007972	2.76	0.014353	4.96	0.016418	5.68	0.987952
S6	1643	0.003730	2.15	0.015707	8.91	0.016100	9.13	0.971994
	21341	0.005257	2.60	0.010682	5.28	0.011905	5.89	0.991371

4.3 ERRORS BARS ON TOC REFLECTANCES

4.3.1 Global statistics

The CGLOPS-1 TOC reflectance product also contains errors bars. They are calculated for each band by propagating the uncertainties on TOA reflectances and ancillary data (only total column of ozone, total column of water vapour, surface pressure and AOT for V1, uncertainty due to the aerosols model is added as of V2 (Section 2.2.2)) to the TOC reflectances. The method to calculate the error bars is described in [CGLOPS1_ATBD_S3-AC-V1.1]. The mean values of the error bars are shown in Figure 30 and Figure 31 for, respectively, OLCI and SLSTR. The mean values of error bars correspond to the same match-ups as used to determinate Accuracy and Uncertainty: SZA smaller than 65 degrees and AOT smaller then 1 (see Section 3.2.1.1). Values of the mean error bars are two to three times smaller than the values of uncertainty for the version 1. The difference between the error bars and the uncertainty for the version 2 is smaller but the errors bars are still smaller than the uncertainty except the bands in the SWIR for SLSTR where the TOA reflectances values are increased by application of significant adjustment calibration factors (more than 10%): that makes confident with the meaning and representativeness of this error bar by taking into account the uncertainty due to the aerosols model. Nevertheless, it is very hard to interpret this comparison with ACIX metrics since the ACIX methodology assumes that there is no error on TOA reflectances.

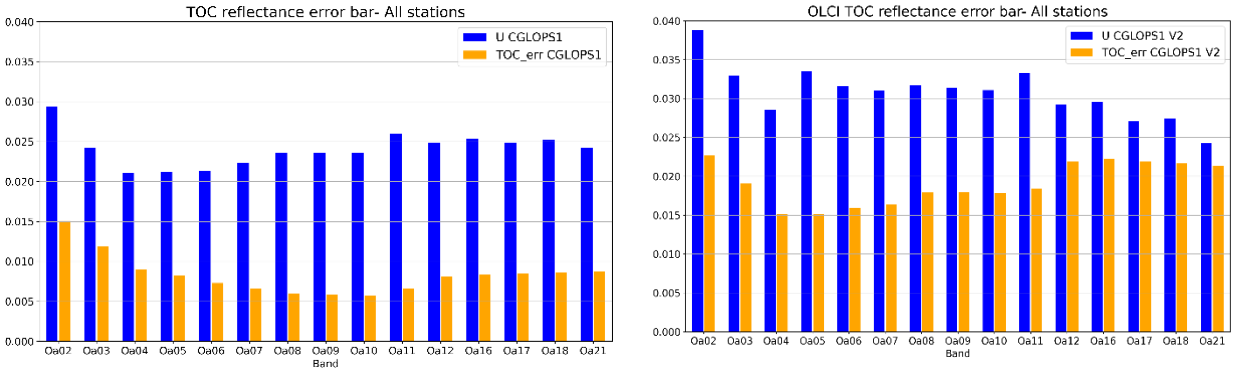


Figure 30: Mean error bars, in yellow, compared to values of Uncertainty (in blue) for the version 1 (left) and the version 2 (right) and for the OLCI bands.

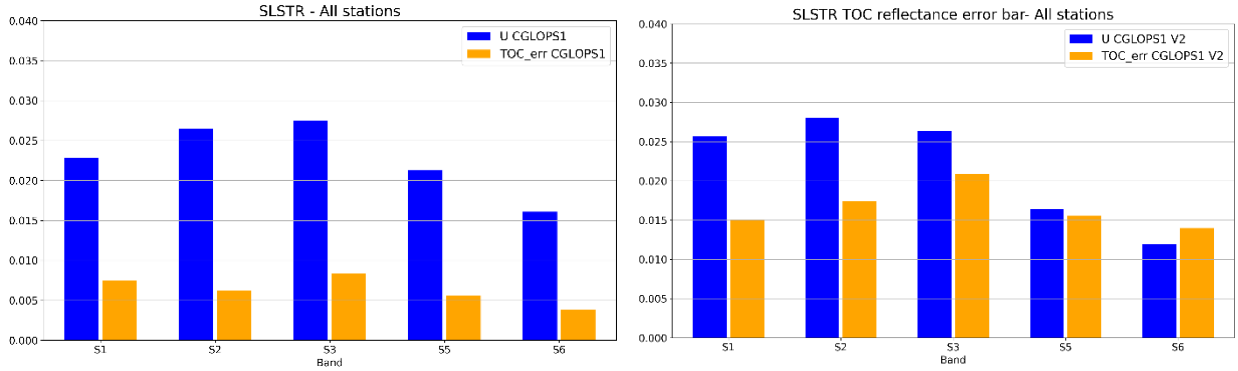


Figure 31: Same as Figure 30 but for SLSTR bands

4.3.2 Evaluation of the error bars for individual stations

To better evaluate the added value of the error bars in the version V2, results can be presented for each individual AERONET sites. The values of the uncertainty and the error bars are analyzed for 4 different bands: 3 bands of OLCI in the different part of the spectrum (Oa3 at 442 nm, Oa8 at 665 nm and Oa17 at 865 nm) and one band of SLSTR in the SWIR (S6 at 2.2 µm). The results are shown in Figure 32 for the band Oa03 de OLCI (443 nm). The results for the other bands are shown in ANNEX 4: Values of errors bars for individual AERONET stations. Two trends can be identified from these plots: (i) a spectral trend and (ii) a differentiation between turbid sites and sites with moderate aerosols load and model variation. For short wavelength, typically in the visible (band Oa03 and Oa08), for low values of the uncertainty, the error bars are in the same order. For stations with large uncertainties, error bar values are significantly smaller than the uncertainties. In the visible the contribution of the atmospheric signal compared to the surface signal is significantly larger than in the near infrared and in the SWIR.

AERONET sites with largest uncertainties are sites often contaminated by urban pollution (big cities as Beijing in China, Dhaka University in Bangladesh, Kanpur in India, Mexico City in Mexico) or sites often contaminated by dust event such as Tamanrasset (Algeria), Banizoumbou (Niger), Dakar (Senegal) Sede Boker (Israel). All these sites can have large, and rapid, variation of AOT and aerosols model that cannot be accurately predicted by CAMS forecast data. It explains that, in the visible, the uncertainty is higher than the error bars. This effect is attenuated in the near infrared and the SWIR because the atmospheric signal is smaller. So, in the near infrared, uncertainties and error bars are in the same order. In the SWIR, it is not as well balanced and error bars tend to be greater than uncertainties.

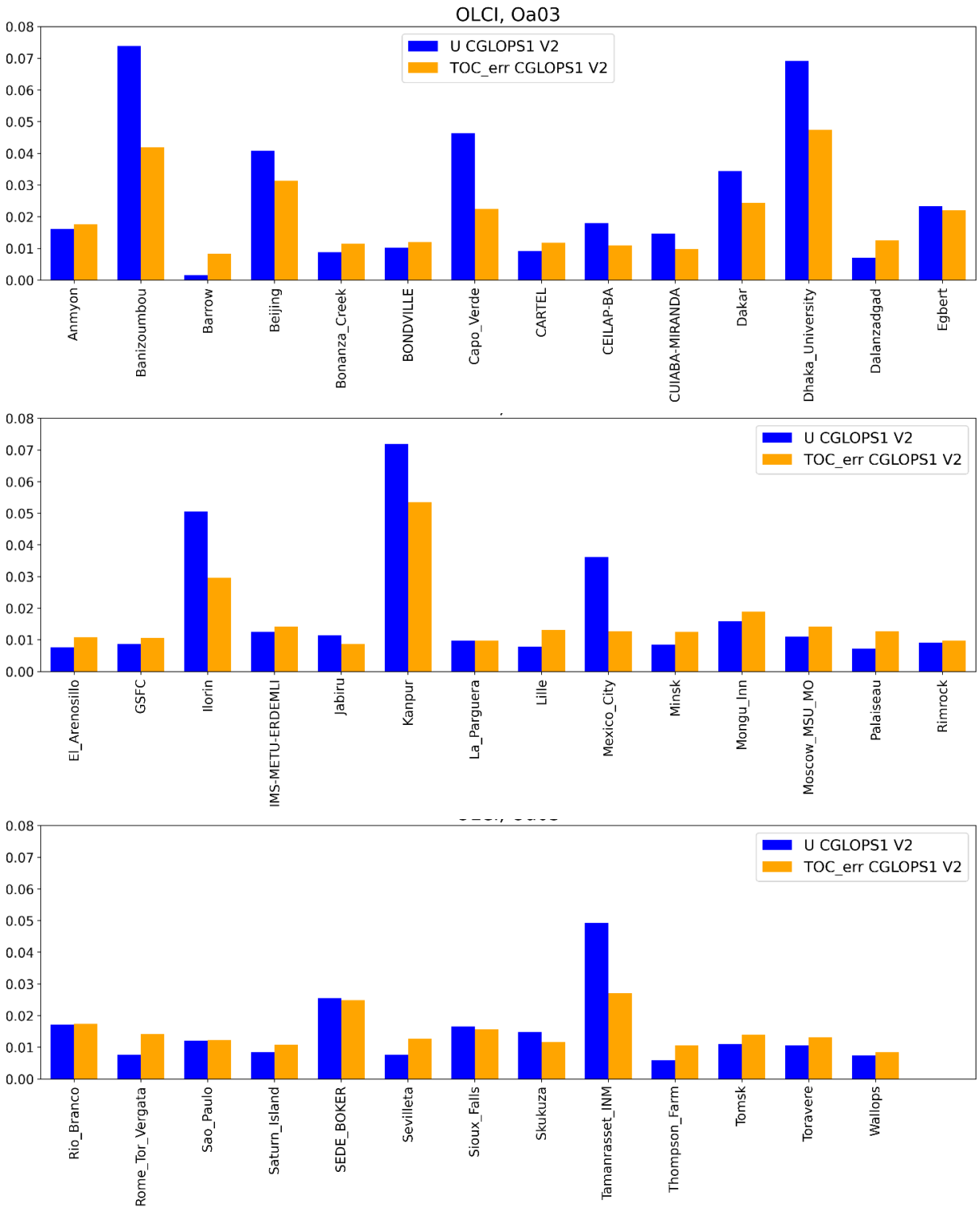


Figure 32: Values of the version 2 TOC reflectance errors bars (in yellow) and the uncertainty (in blue) for the OLCI band OA03 (443 nm) for the 41 AERONET stations

The size of the dataset of each AERONET stations is given in Figure 33. By size of the dataset, we understand the number of TOC reflectances obtained for clear sky conditions with an AOT less than 1 and a SZA smaller than 65°. The graphs clearly indicate that some stations have large weight compared to some others. It can be explained by the low cloud contamination for these stations. That is the case for Tamanrasset (Algeria), Banizoumbou (Niger), Dalanzagbad (Mongolia), El Arenosillo (Spain) or Sede Boker (Israel). About 30% of the whole dataset (including all AERONET sites) are from Banizoumbou and Tamanrasset that are the sites with the worse quality for the accuracy, the precision and the uncertainty.

In conclusion, for AERONET sites and areas with moderate variation of the atmospheric and aerosols properties, it is valuable that errors bars and uncertainties are comparable: it clearly means that we well take into account the different source of uncertainty on atmospheric composition. For turbid areas, it is hard to really predict the error bars on the TOC reflectance because forecast ancillary data are used for aerosols. It is especially important in the visible because the atmospheric contribution compared to the measured TOA signal is very large.

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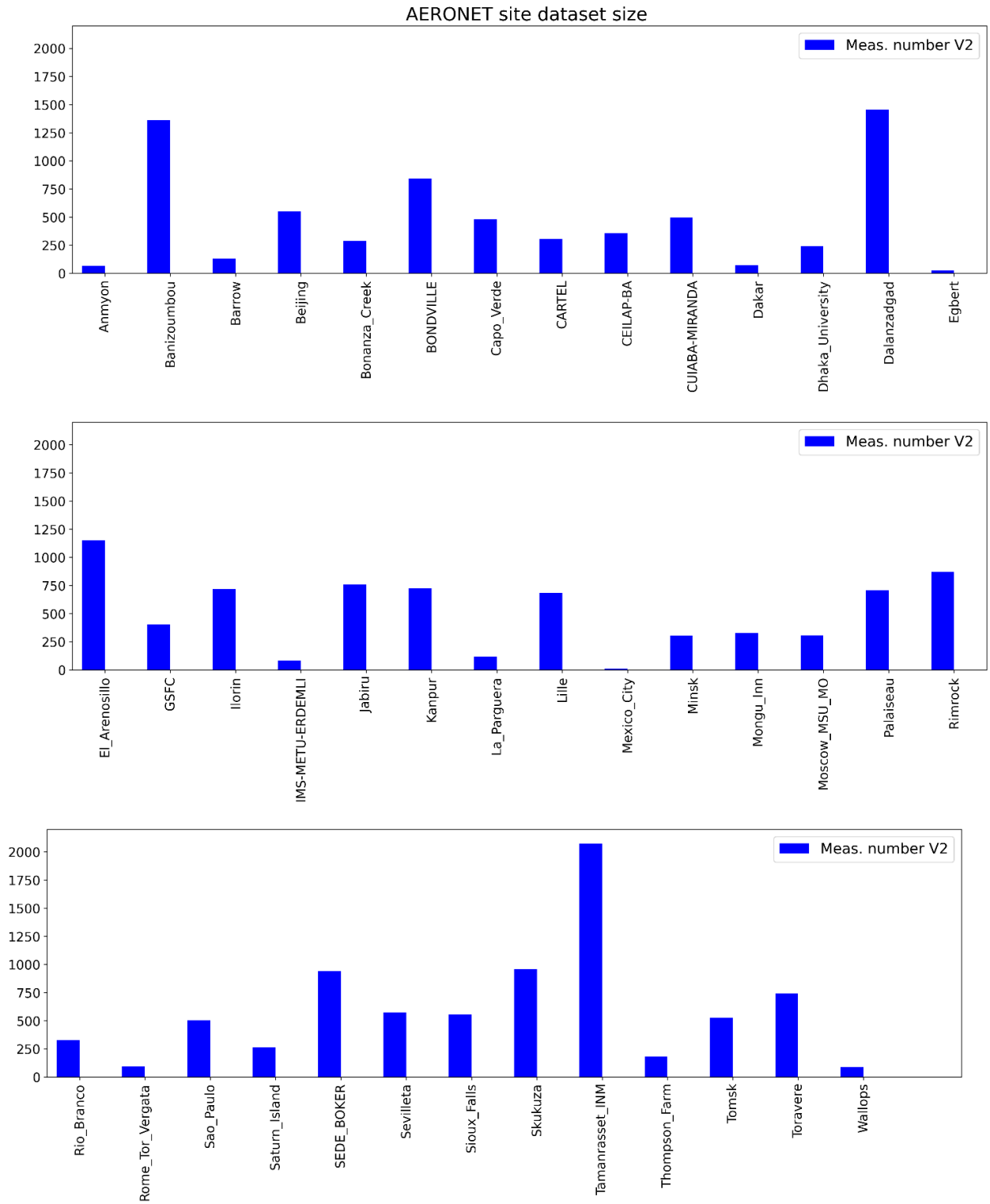


Figure 33 : Number of available Sentinel-3 measurements for all AERONET stations involved in the validation.

5 CONCLUSIONS, RECOMMENDATIONS AND PERSPECTIVES

The Simplified Model for Atmospheric Corrections (SMAC) algorithm (Rahman and Dedieu, 1994) is used to perform atmospheric correction in the TOA-to-TOC module of the CGLS Sentinel-3 OLCI and SLSTR biophysical processing chains. Inputs like the Aerosol Optical Thickness (AOT), the total column of ozone, the amount of water vapour and the surface pressure are from ancillary data. The aerosol model is assumed to be the "continental" one.

The atmospheric correction scheme applied on Sentinel-3 OLCI and SLSTR TOA reflectances has been performed for three sources of ancillary data: daily CAMS NRT products at 3-hours resolution, daily MERRA-2 reanalysis at 1-hour resolution, and a monthly synthesis of MERRA-2 as a climatology. For the operational processing, daily CAMS NRT products are used because MERRA-2 reanalysis is not available in near real time. The climatology is considered as a back-up.

AOT from the ancillary data have been compared to AOT from AERONET inversion products Version 3. The best agreement between AERONET and ancillary data is found for MERRA-2 reanalysis with a mean bias of -0.0098 and a standard deviation of 0.1182. For CAMS NRT and the climatology, the mean bias is respectively 0.0183 (±0.1172) and 0.0359 (±0.1400). No values of AOT above 1.0 is found for the climatology: this source of ancillary data is not able to be representative of extreme aerosols events. CAMS NRT products have a trend to give high AOT values not found in AERONET datasets.

To assess the performances of the AC, we have followed the ACIX methodology (Doxani et al, 2018). The Top of canopy reflectances (TOC-r) have been compared with a reference product obtained from AERONET data and the Second Simulation of the Satellite Signal in the Solar Spectrum (6S) accurate atmospheric correction method. 48 AERONET sites have been chosen to cover different meteorological and typical atmospheric conditions, and different types of surfaces. One year of S3-A (July 2018 – June 2019) and 6 months of S3B data (January 2019 – June 2019) have been considered. The metrics of ACIX are used to quantify the performance. The accuracy A is the mean bias between the AC processor TOC-r and the reference product, the precision P is the standard deviation of the difference and the uncertainty U is the root mean square deviation.

Filtering pixels contaminated by clouds or their shadows by the use of the cloud mask from SLSTR and the pixel classification flag from OLCI, more than 1700 match-ups were found. A best case has been defined, keeping pixel with moderate solar zenith angles, less than 65 degrees, and value of AOT smaller than 1.0, (i) to avoid conditions of the limit of the SMAC method based on fit of radiative transfer simulations and (ii) to remove bad conditions as the validity of the plane-parallel assumption in radiative transfer code and to remove cases with high atmospheric contribution compared to the surface.

A spectral dependency of the accuracy (from a few 10^{-3} to 1.2 10^{-2}) is obtained. CAMS NRT is better than MERRA-2 for shorter wavelengths (< ~720nm), but MERRA-2 is better at longer wavelengths (> ~720 nm). Climatology shows results a little degraded compared to the two others sources of ancillary data. Precision and uncertainty are between 0.012 and 0.033 for CAMS and MERRA-2. MERRA-2 and CAMS NRT have very closed values of precision and uncertainty. Climatology gives the worse results (around 20 % more). Finally, the uncertainty is mainly driven by the dispersion not the bias. These results are summarized in Annex 3: Statistics of Surface Reflectances Comparison.

If we extend the analysis to AOT greater than 1.0, accuracy, precision and accuracy are degraded. The performances of SMAC AC scheme are not acceptable anymore for large amounts of aerosols. For grazing solar angle, i. e. for SZA greater than 65, a completely different trend on ACIX metrics is identified. The mean bias, i. e. the accuracy, is negative for bands at short wavelengths but is almost equal in absolute values. Its value is around zero for longer wavelengths. Precision and uncertainty are smaller than those obtained for the best conditions. TOC reflectances can then be used for high values of SZA.

The SMAC processor using CAMS NRT ancillary data for atmospheric parameters is suitable for the generation of the Sentinel-3 TOC reflectances. However, we recommend excluding pixels with an AOT greater than 1.0 for further processing. TOC reflectances calculated for large SZA greater than 65° can be used to cover areas for several weeks during winter time, but they must be handled with care because of the modelling of the Earth-Atmosphere system in radiative transfer code (plane-parallel horizontally homogenous layers). We recommend to keep this information in the quality flag of the biophysical products.

The Climatology is used as back-up in case of CAMS NRT are not available in due time for the processing. The TOC reflectance will be slightly degraded; so, it is important to keep in mind this information for the use of the TOC reflectances.

In the AC scheme, the aerosols model is assumed to be the “continental” model, what is source of uncertainty and systematic bias. The two versions of the processing line that calculates TOC reflectances and their errors bars has been compared. The algorithm to calculate the TOC reflectances is exactly the same for OLCI data whereas adjustment factors of calibration, less than a few percents in the visible and the NIR but greater than 10% in the SWIR, are applied to SLSTR TOA reflectances. Moreover, the uncertainty due to aerosols model is included in the calculation of the error bars of TOC reflectance as of version 2.0. Errors bars of version 2.0 are comparable to the value of the ACIX uncertainty, except for AERONET stations with large and rapid variability of AOT and aerosols model such as areas often affected by dust events or urban pollution. The error bars of version 2.0 are then useful because providing realistic values. As explained in Section 4.1, **the results obtained for version 2.0 are valid for the TOC reflectance version 2.3**.

A possible evolution could have been to select a model from CAMS NRT aerosol optical thickness given for 5 components (sea salt, dust, sulphate, organic matter and black carbon) and the OPAC database (Hess et al., 1998). This possibility has been tested during preparation of the Collection 2 of PROBA-V and does not improve the TOC reflectances due, partly, to sometimes large differences between AERONET and CAMS-NRT. An explanation may be the difficulty to link aerosols optical properties with fraction of optical thickness for chemical components.

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ANNEX 1: SPECTRAL BANDS OF SENTINEL-3 SENSORS

Table 4: S3A OLCI spectral bands. The third column, so-called "corrected", indicates if the TOA signal in the band is corrected from atmospheric effect and thus used in the processing.

Band	Central wavelength (nm)	FWHM (nm)	Corrected	Absorbing gas (transmission)*
Oa01	399.9	14.0	no	O_3 (0.996)
Oa02	411.8	9.8	yes	O_3 (0.996)
Oa03	442.9	9.9	yes	O_3 (0.995)
Oa04	490.4	9.9	yes	O_3 (0.990)
Oa05	510.4	9.9	yes	O_3 (0.983) ; H_2O (0.997)
Oa06	560.4	9.9	yes	O_3 (0.961)
Oa07	620.4	9.9	yes	O_3 (0.960)
Oa08	665.2	9.9	yes	O_3 (0.981) ; H_2O (0.998)
Oa09	674.0	7.4	yes	O_3 (0.985)
Oa10	681.6	7.4	yes	O_3 (0.987)
Oa11	709.1	9.9	yes	O_3 (0.993) ; H_2O (0.971)
Oa12	754.2	7.4	yes	O_3 (0.997)
Oa13	761.7	2.5	no [1]	O_3 (0.997) ; CO_2 (0.985) ; O_2 (0.333)
Oa14	764.8	3.7	no [1]	O_3 (0.998) ; CO_2 (0.990) ; O_2 (0.532)
Oa15	767.9	2.5	no [1]	O_3 (0.998) ; O_2 (0.875)
Oa16	779.2	15.0	yes	O_3 (0.997) ; H_2O (0.998)
Oa17	865.6	19.8	yes	H_2O (0.998)
Oa18	884.4	9.9	yes	H_2O (0.992)
Oa19	899.4	9.9	no [2]	H_2O (0.588)
Oa20	939.3	19.7	no [2]	H_2O (0.096)
Oa21	1 012.9	26.9	yes	H_2O (0.988)

(*) typical gaseous transmission for a unit air mass and a US standard atmosphere. Transmissions greater than 0.999 are not reported.

[1] This band is highly contaminated by oxygen absorption (O_2)

[2] This band is highly contaminated by Water Vapour (WV)

Similarly, for S3B OLCI, bands Oa01, Oa13, Oa14, Oa15, Oa19 and Oa20 are not corrected from atmospheric effects.

Table 5: same as Table 4 but for S3A SLSTR

Band	Central wavelength (nm)	FWHM (nm)	Corrected	Absorbing gas (transmission)*
S1	554.3	19.2	yes	O_3 (0.965); H_2O (0.999)
S2	659.5	19.2	yes	O_3 (0.979); H_2O (0.986)
S3	868.0	20.6	yes	H_2O (0.999)
S4	1 374.8	20.4	No [1]	H_2O (0.000); CH_4 (0.997)
S5	1 613.4	66.4	yes	CO_2 (0.978); H_2O (0.995); CH_4 (0.996)
S6	2 255.5	51.0	yes	H_2O (0.991); CH_4 (0.944); N_2O (0.996)

(*) typical gaseous transmission for a unit air mass and a US standard atmosphere. Transmissions greater than 0.999 are not reported.

[1] This band is used for the detection of cirrus.

Similarly, for S3B SLSTR, band S4 is not corrected from atmospheric effects.

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ANNEX 2: AERONET SITES

Table 6: List of AERONET sites selected for the validation. Latitude, longitude and country are given. Grey sites are not included in the analysis of TOC-r (Section 4.2)

Site	Latitude (N°)	Longitude (°E)	Country	Site	Latitude (°N)	Longitude (°E)	Country
Anmyon	36.53	126.33	South Korea	La_Parguera	17.97	-67.04	Puerto Rico
Arica	-18.47	70.31	Chile	Lille	50.61	3.14	France
Banizoumbou	13.54	2.66	Niger	Mexico_City	19.33	-99.18	Mexico
Barrow	71.31	-156.66	Alaska (US)	Minsk	53.92	27.6	Belarus
Beijing	39.97	116.38	China	Mongu_Inn	-15.26	23.13	Zambia
Bonanza_Creek	64.74	-148.31	Alaska (US)	Moscow_MSU_MO	55.70	37.52	Russia
Bondville	40.05	-88.37	Illinois (US)	Palaiseau	48.71	2.21	France
Capo_Verde	16.73	-22.93	Sal Island (Capo Verde)	Rimrock	46.48	-116.99	Idaho (US)
Carpentras	44.08	5.05	France	Rio_Branco	-9.95	-67.86	Brazil
Cartel	45.37	-71.93	Canada	Rome_Tor_Vergata	41.83	12.64	Italy
Cart_Site	36.60	-97.48	Oklahoma (US)	Sao_Paulo	-23.56	-46.73	Brazil
Ceilap-BA	-34.55	-58.5	Argentina	Saturn_Island	48.77	-123.12	Canada
Cuiaba-Miranda	-15.73	-56.07	Brasil	Sede_Boker	30.85	34.78	Israel
Dakar	14.39	-16.95	Senegal	Sevilleta	34.35	-106.88	New Mexico (US)
Dhaka_University	23.72	90.39	Bangladesh	Shirahama	33.69	135.35	Japan
Dalanzadgad	43.57	104.41	Mongolia	Sioux_Falls	43.73	-96.62	South Dakota (US)
Egbert	44.23	-79.78	Canada	Skukuza	-24.99	31.58	South Africa
El_Arenosillo	37.10	-6.73	Spain	Tamanrasset_INM	22.79	5.53	Algeria
Forth_Crete	35.33	25.28	Greece	Thompson_Farm	43.10	-70.94	New Hampshire (US)
GSFC	38.99	-76.83	Maryland (US)	Tomsk	56.47	85.04	Russia
Ilorin	8.48	4.67	Nigeria	Toravere	58.26	26.46	Estonia
IMS-METU-Erde-mli	36.56	34.25	Turkey	Venise	45.31	12.50	Italy
Jabiru	-12.66	132.89	Australia	Wallops	37.93	-75.47	Virginia (US)
Kanpur	26.51	80.23	India				

ANNEX 3: STATISTICS OF SURFACE REFLECTANCES COMPARISON.

Table 7: Statistics of the comparison between TOC reflectances calculated with the SMAC processor using CAMS NRT ancillary data and 6S using AERONET inversion products for each band of OLCI and SLSTR. N is the number of match-ups. A is the accuracy (and its relative value in %), P is the precision (and its relative value in %), U the uncertainty (and its relative value in %) and R2 is the correlation factor. Only cloud free pixels, with a SZA smaller than 65° and AOT smaller than 1.0 are kept

Band	N	A CAMS	A (%)	P CAMS	P (%)	U CAMS	U (%)	R2 CAMS
Oa02	1705	-0.000289	-0.44	0.029270	44.29	0.029271	44.29	0.406619
Oa03	1712	0.000904	1.13	0.024154	30.30	0.024171	30.32	0.656674
Oa04	1715	0.002688	2.87	0.020878	22.28	0.021050	22.46	0.812406
Oa05	1718	0.003953	3.86	0.020741	20.26	0.021114	20.62	0.842843
Oa06	1720	0.005182	3.87	0.020649	15.43	0.021289	15.91	0.904127
Oa07	1720	0.005913	3.95	0.021469	14.34	0.022268	14.87	0.944263
Oa08	1720	0.006744	4.30	0.022559	14.38	0.023545	15.01	0.950677
Oa09	1720	0.006614	4.19	0.022592	14.30	0.023540	14.90	0.952306
Oa10	1720	0.006600	4.14	0.022554	14.13	0.023500	14.72	0.953295
Oa11	1720	0.009437	4.75	0.024077	12.12	0.025860	13.01	0.934038
Oa12	1720	0.005976	2.16	0.024013	8.66	0.024745	8.93	0.924586
Oa16	1720	0.007837	2.73	0.024064	8.38	0.025308	8.82	0.926677
Oa17	1720	0.006345	2.05	0.023956	7.76	0.024782	8.02	0.930894
Oa18	1720	0.007098	2.27	0.024079	7.70	0.025103	8.02	0.930893
Oa21	1720	0.004609	1.37	0.023692	7.04	0.024136	7.17	0.940598
S1	1720	0.005030	3.62	0.020658	14.88	0.021262	15.32	0.910924
S2	1720	0.007028	4.35	0.022848	14.13	0.023905	14.78	0.951746
S3	1720	0.006238	2.01	0.024160	7.80	0.024952	8.06	0.938903
S5	1701	0.005342	2.10	0.016770	6.58	0.017601	6.91	0.975170
S6	1701	0.003656	2.09	0.011616	6.63	0.012177	6.95	0.984544

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Table 8: same as Table 7 but for MERRA2 reanalysis

Band	N	A MERRA-2	A (%)	P MERRA-2	P (%)	U MERRA-2	U (%)	R^2 MERRA-2
Oa02	1703	0.002753	3.99	0.033453	48.53	0.033566	48.69	0.341791
Oa03	1709	0.003420	4.17	0.028178	34.37	0.028385	34.63	0.581512
Oa04	1709	0.004570	4.79	0.024288	25.47	0.024714	25.92	0.759589
Oa05	1710	0.005595	5.39	0.023739	22.87	0.024389	23.50	0.801865
Oa06	1710	0.006149	4.58	0.023262	17.31	0.024061	17.91	0.880376
Oa07	1710	0.006538	4.36	0.023780	15.87	0.024662	16.46	0.932147
Oa08	1710	0.007130	4.54	0.024610	15.68	0.025622	16.33	0.941597
Oa09	1710	0.006952	4.40	0.024596	15.56	0.025560	16.17	0.943732
Oa10	1710	0.006914	4.33	0.024514	15.35	0.025471	15.95	0.945076
Oa11	1710	0.009871	4.97	0.025804	13.00	0.027627	13.92	0.925540
Oa12	1710	0.005084	1.84	0.025667	9.29	0.026165	9.47	0.915290
Oa16	1710	0.006845	2.39	0.025570	8.94	0.026471	9.26	0.918581
Oa17	1710	0.005236	1.70	0.025082	8.15	0.025623	8.33	0.925337
Oa18	1710	0.006127	1.97	0.025121	8.06	0.025858	8.30	0.925943
Oa21	1710	0.003837	1.15	0.024441	7.30	0.024741	7.39	0.937713
S1	1710	0.005943	4.26	0.023432	16.82	0.024174	17.35	0.887107
S2	1710	0.007580	4.69	0.025005	15.46	0.026129	16.15	0.942450
S3	1710	0.005103	1.65	0.025329	8.21	0.025838	8.38	0.933485
S5	1691	0.004852	1.91	0.017002	6.70	0.017681	6.97	0.974513
S6	1691	0.003409	1.95	0.011445	6.55	0.011942	6.84	0.984994

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Table 9: Same as Table 7 but for the Climatology

Band	N	A Clim	A (%)	P clim	P (%)	U Clim	U (%)	R^2 Clim
Oa02	1700	0.001193	1.76	0.043487	64.28	0.043503	64.31	0.194617
Oa03	1717	0.001808	2.24	0.036100	44.77	0.036145	44.83	0.433714
Oa04	1720	0.003347	3.55	0.030000	31.79	0.030187	31.99	0.664146
Oa05	1722	0.004769	4.62	0.028577	27.69	0.028973	28.07	0.730162
Oa06	1725	0.006090	4.52	0.026205	19.46	0.026903	19.98	0.852785
Oa07	1723	0.007032	4.66	0.025465	16.88	0.026419	17.51	0.924129
Oa08	1723	0.007891	4.99	0.025774	16.31	0.026955	17.06	0.937701
Oa09	1723	0.007701	4.84	0.025687	16.15	0.026816	16.86	0.940321
Oa10	1723	0.007687	4.78	0.025542	15.89	0.026674	16.60	0.942042
Oa11	1725	0.012107	6.02	0.026346	13.09	0.028994	14.41	0.924943
Oa12	1725	0.007540	2.71	0.025862	9.29	0.026939	9.67	0.915509
Oa16	1725	0.009492	3.29	0.025701	8.91	0.027398	9.50	0.919293
Oa17	1725	0.008005	2.58	0.025110	8.09	0.026355	8.49	0.926508
Oa18	1725	0.009141	2.91	0.025087	7.97	0.026700	8.49	0.927672
Oa21	1725	0.006749	1.99	0.024275	7.17	0.025196	7.44	0.939978
S1	1725	0.005987	4.28	0.026266	18.80	0.026940	19.28	0.862658
S2	1724	0.008561	5.25	0.026068	15.97	0.027438	16.81	0.939280
S3	1725	0.007893	2.54	0.025311	8.13	0.026514	8.52	0.934805
S5	1706	0.006283	2.46	0.016959	6.63	0.018085	7.08	0.975393
S6	1706	0.004283	2.44	0.011611	6.61	0.012376	7.04	0.984850

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ANNEX 4: VALUES OF ERRORS BARS FOR INDIVIDUAL AERONET STATIONS

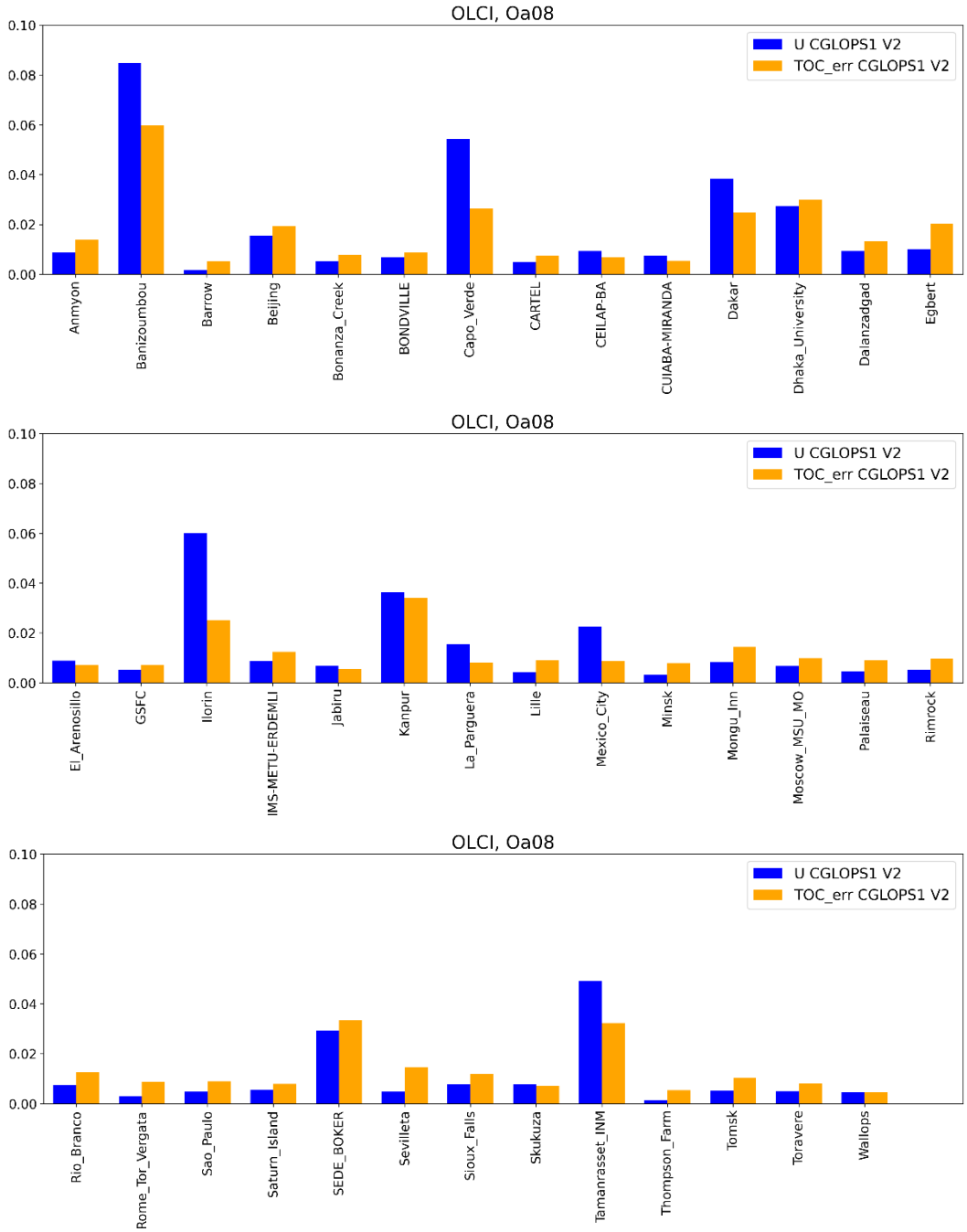


Figure 34: Version 2.0 TOC reflectances error bars (in orange) and uncertainty (in blue) for the OLCI band OA08 (670 nm) for the 41 AERONET sites

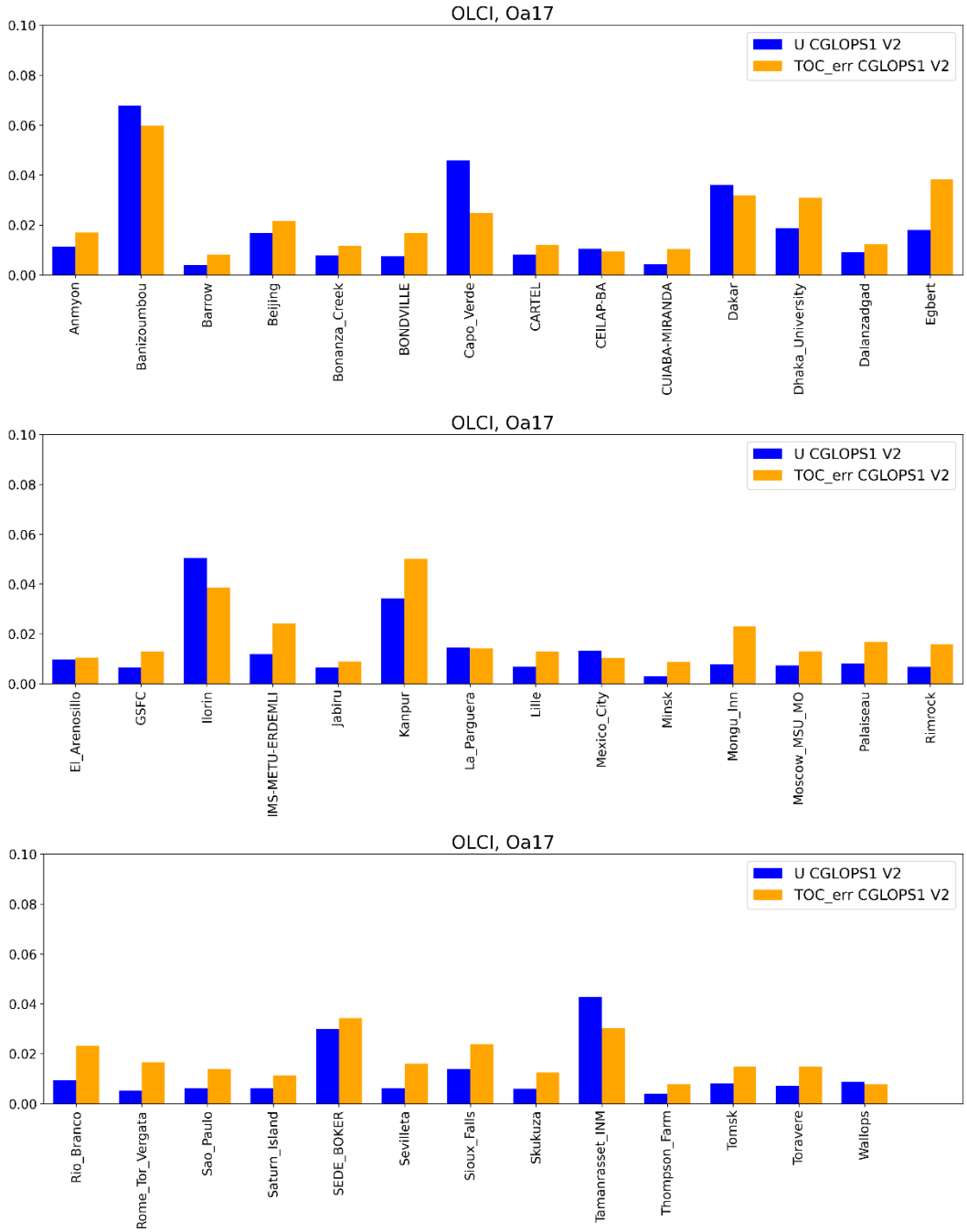


Figure 35: Version 2.0 TOC reflectances error bars (in orange) and uncertainty (in blue) for the OLCI band OA17 (865 nm) for the 41 AERONET sites

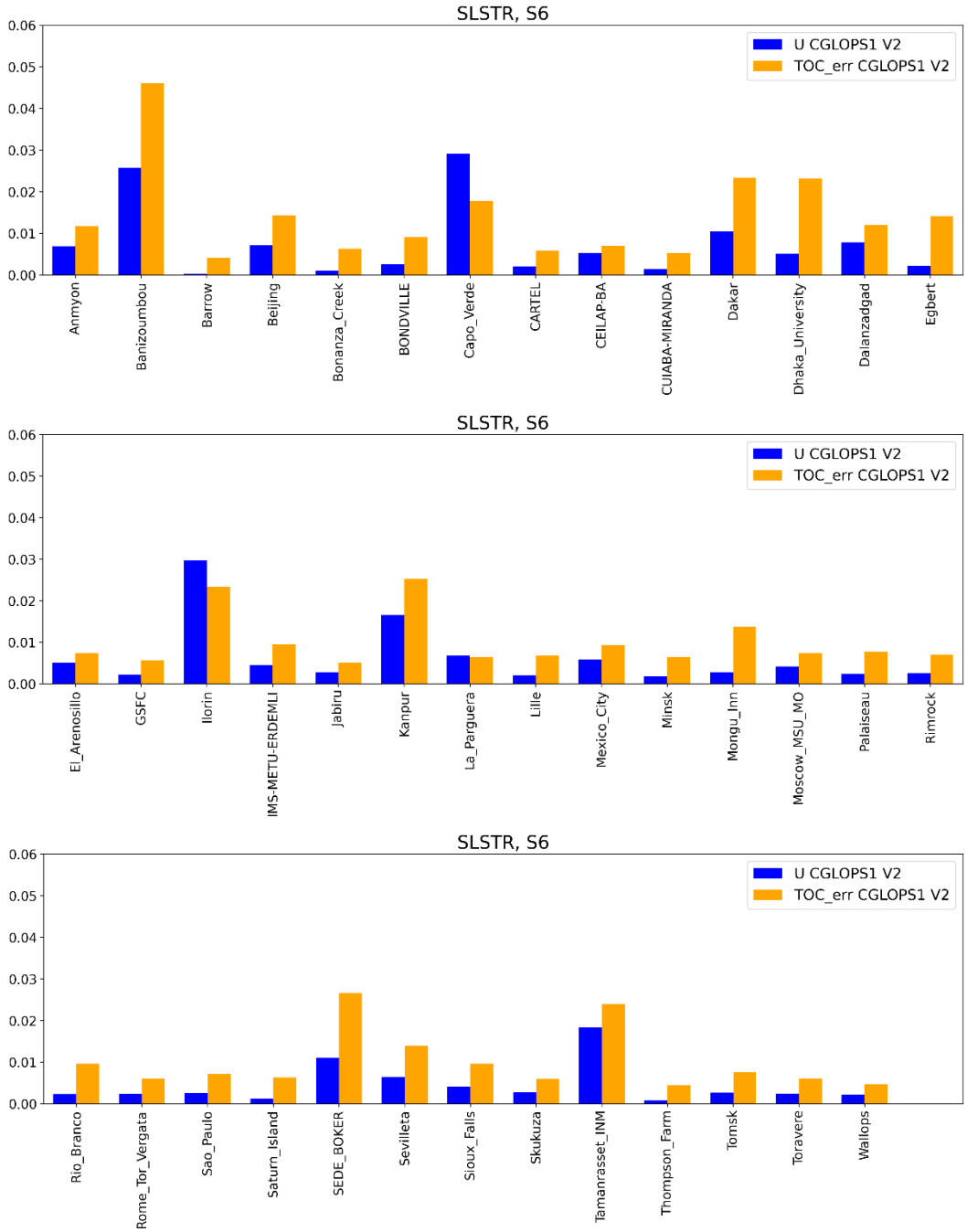


Figure 36: Version 2.0 TOC reflectances error bars (in orange) and uncertainty (in blue) for the SLSTR band S6 (2255 nm) for the 41 AERONET sites